Understanding Q-Learning and Deep Q-Learning in 2026: A Methodological and Empirical Survey

Anonymous Submission to IEEE Transactions on Artificial Intelligence

***Abstract*—Q-learning remains a cornerstone of Reinforcement Learning (RL), underpinning many state-of-the-art deep RL algorithms. Yet, despite its foundational status, no survey has systematically organized the diverse innovations that have shaped Q-learning over the past three decades. This paper fills that gap by presenting a comprehensive review of Q-learning variants, organized around the eight foundational weaknesses of vanilla Q-learning that modern methods address — overestimation bias, sample inefficiency, brittle exploration, reward sparsity and credit assignment, distribution shift, multi-agent coordination, slow adaptation, and function-approximation instability. We introduce a unified taxonomy covering both tabular and deep Q-learning methods, including offline RL, multi-agent value decomposition, and distributed Q-learning, and compile benchmark results from original papers across Atari and classic control environments. To complement the literature review, we provide a comparative analysis of six major open-source deep RL repositories — Tianshou, XuanCe, CleanRL, DQN Zoo, Stable Baselines3, and RLlib — highlighting their coverage of Q-learning algorithms and practical implementation challenges. Together, these contributions offer a structured resource for researchers and practitioners to navigate the evolution, empirical performance, and implementation landscape of Q-learning.**

***Index Terms*—Deep Learning, Machine Learning, Reinforcement Learning, Q-Learning, Survey**

Impact Statement

Reinforcement learning increasingly underpins decision-making systems in robotics, autonomous vehicles, recommendation, and the post-training of large language models — and value-based Q-learning sits beneath much of this technology. Yet practitioners face a fragmented literature: dozens of algorithmic variants whose purpose, trade-offs, and software availability are scattered across three decades of papers. This survey reorganizes that landscape around the specific weaknesses each method was designed to fix, letting an engineer move directly from a symptom — overestimation, sparse rewards, instability at scale — to the methods that address it and the open-source libraries that implement them. By pairing this diagnostic taxonomy with calibrated benchmark evidence and a reproducibility audit of six widely used codebases, the work lowers the barrier to selecting, deploying, and extending Q-learning methods, and surfaces where research and tooling investment are most needed.

Double-anonymous review copy. Author identities, affiliations, and acknowledgements are withheld and provided via the submission portal.

I. Introduction

Reinforcement Learning (RL) and Deep Reinforcement Learning (DRL) have emerged as powerful paradigms for solving complex sequential decision-making problems, with applications ranging from Atari games [1], [2] and robotic control [3] to autonomous driving [4]. Among DRL approaches, two major families dominate: value-based methods, which estimate action-value functions, and policy-based methods, which directly optimize policy distributions.

While policy-based algorithms have shown strong performance in tasks with continuous action spaces, they often suffer from high variance and sensitivity to hyperparameters. In contrast, value-based methods such as Q-learning [5] and its deep variant DQN [1] remain widely used due to their conceptual simplicity, sample efficiency in discrete domains, and ease of implementation.

Q-learning's longevity has produced a sprawling methodological literature. Over three decades, dozens of variants have been proposed to address specific limitations — overestimation bias, brittle exploration, sample inefficiency, instability under function approximation — and many have been combined into composite agents such as Rainbow [6]. Despite this volume of work, the *organizing structure* through which the field is presented has remained largely chronological or method-typed: distributional methods, ensemble methods, replay innovations, and so on. Existing surveys [7–10], as well as more recent broad-RL treatments [11], have inherited this method-type organization.

Method-type organization is bibliometrically convenient but analytically thin. It tells the reader *what kind of thing* a method is (an ensemble, a distributional model, a replay buffer modification) without surfacing *why* the method exists — which weakness of vanilla Q-learning it was designed to address, what trade-offs it introduces, and how it compares to other approaches that target the same weakness. As a result, readers encounter algorithms as isolated artifacts rather than as positioned moves in an ongoing technical conversation. Cross-method synthesis suffers accordingly.

This paper offers an alternative organization. Rather than presenting Q-learning's variants by method type, we organize the field around **the foundational weaknesses of vanilla Q-learning that modern methods address**. Eight such weaknesses are identified:

1. *Overestimation bias* introduced by the `max` operator over noisy value estimates;

2. *Sample inefficiency* arising from uniform experience replay;
3. *Brittle exploration* under ε-greedy action selection;
4. *Reward sparsity and credit assignment* in long-horizon tasks;
5. *Distribution shift* when transferring to fixed-data (offline) regimes;
6. *Multi-agent coordination* failures when factoring single-agent Q across cooperative agents;
7. *Slow adaptation* when training per-task without transfer; and
8. *Function-approximation instability* — the "deadly triad" of off-policy learning, bootstrapping, and function approximation.

Each Related Works section in this paper corresponds to one of these weaknesses. Within each section, methods are grouped by the *mechanism* they use to address the weakness, compared head-to-head on the trade-offs they introduce, and evaluated against the empirical evidence relevant to that axis. The same method can appear in multiple sections when it advances multiple axes — Rainbow [6], for instance, recurs across five of the eight — and these cross-references form an explicit map of the field's connective tissue.

This organization preserves all of the paper's empirical and bibliographic contributions while reframing the surrounding analysis. Five distinguishing contributions, summarized in Table I, support the reframing:

1. A unified problem-axis taxonomy that integrates tabular Q-learning, classical deep Q-learning, and modern Q-based methods (offline RL, multi-agent value decomposition, distributed scaling) within a single framework.
2. A comparative analysis of six widely used open-source DRL repositories — Tianshou [12], XuanCe [13], CleanRL [14], DQN Zoo [15], Stable Baselines3 [16], and RLlib [17] — annotated *by axis* to reveal where the open-source ecosystem provides coverage and where it does not.
3. Curated benchmark results extracted from the Atari [2] suite as reported in the original papers, stratified by task category, and re-interpreted as evidence for or against each algorithmic axis rather than as a leaderboard.
4. Tabular Q-learning benchmark results from our own controlled reimplementations, isolating algorithmic design from architectural confound.
5. A thorough per-paper literature review organized by axis, with each method positioned by the weakness it addresses, the mechanism it uses, and the trade-offs it introduces.

The remainder of the paper is organized as follows. Section II introduces the MDP formalism and formally states the eight weaknesses that structure the Related Works. Section III describes the methodology. Section IV — the bulk of the paper — presents the problem-first review across the eight axes (§IV.A–H), followed by a ninth subsection §IV.I covering recent theoretical advances. Sections V and VI report Atari and tabular benchmark analyses. Section VII presents the repository comparison. Section VIII concludes with future directions, including a planned community-maintained Q-learning repository whose roadmap is informed by the gaps identified in Section VII. Supplementary material provides the full per-method mapping between the problem-first axis taxonomy of §IV and the method-type taxonomy of prior surveys, together with notation conventions and selected derivations.

Reading guide. The §IV body is written for mechanism and trade-off: each axis-section names the weakness, surveys the solution families that respond, and compares them on the costs they introduce. Equations in §IV are mechanism-defining rather than fully derived; longer derivations and proof sketches are concentrated in the supplementary material and may be skipped on a first read without losing the structural argument. §IV.I revisits the same ground theoretically.

II. Background and Problem Setup

II.A. Markov Decision Processes: We consider the standard formulation of a Markov Decision Process (MDP), defined as $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, R, P, \gamma, d_0 \rangle$, where $\mathcal{S}$ denotes the set of all possible states, and $\mathcal{A}$ the set of possible actions. The reward function $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ assigns a real-valued reward to each state-action pair. The transition function $P(s' \mid s, a)$ specifies the probability of transitioning to state s' from state s after taking action a. The discount factor $\gamma \in [0, 1)$ determines the importance of future rewards, and d_0 denotes the initial state distribution.

A policy $\pi : \mathcal{S} \to \mathcal{A}$ (or, in the stochastic case, $\pi : \mathcal{S} \to \Delta(\mathcal{A})$) prescribes an action for each state. The value of a policy at state s is the expected discounted return under that policy: $V^\pi(s) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s\right]$. The associated action-value function is $Q^\pi(s, a) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s, a_0 = a\right]$. Q-learning [5] estimates the optimal action-value $Q^*(s, a) = \max_\pi Q^\pi(s, a)$ via the recursive Bellman optimality equation,

$$Q^*(s, a) = \mathbb{E}_{s' \sim P(\cdot|s,a)}\left[r + \gamma \max_{a'} Q^*(s', a')\right],$$

implemented as an iterative update,

$$Q(s, a) \leftarrow Q(s, a) + \alpha\Big(r + \gamma \max_{a'} Q(s', a') - Q(s, a)\Big).$$

The remainder of this paper surveys methods that modify, augment, or replace components of this update to address one or more of the weaknesses introduced below.

II.B. Eight Weaknesses of Vanilla Q-Learning: The Q-learning update of §II.A is provably convergent under tabular representation and standard stochastic approximation conditions [Watkins & Dayan 1992]. In practical settings — function approximation, partial coverage, sparse rewards, multi-agent environments, fixed-data regimes — the update exhibits eight distinct failure modes. These failure modes are not independent; several share underlying mechanisms. But each motivates a recognizably different family of methods, and each is the organizing principle of one Related Works subsection.

W1. Overestimation bias. The operator $\max_{a'} Q(s', a')$ is biased upward in expectation whenever $Q(s', \cdot)$ contains zero-mean noise. Formally, for noisy estimators $\tilde{Q}(s', a) = Q^*(s', a) + \epsilon_a$ with $\mathbb{E}[\epsilon_a] = 0$,

$$\mathbb{E}\Big[\max_{a'} \tilde{Q}(s',a')\Big] \geq \max_{a'} Q^*(s',a'),$$

with strict inequality whenever the noise is not degenerate [Thrun & Schwartz 1993]. Under recursive Bellman backups this bias propagates and amplifies, steering the greedy policy toward actions whose values are least accurately estimated. Methods responding to this weakness are surveyed in §IV.A.

W2. Sample inefficiency. Vanilla Q-learning uses each transition once and uniformly. The introduction of experience replay [18] allowed transitions to be reused, but uniform sampling treats all transitions as equally informative — a transition where the agent already predicts the outcome accurately contributes little to the update, yet is sampled as often as a high-error transition. In environments where high-information transitions are rare (sparse reward, low-probability state encounters), uniform replay yields slow convergence. Methods responding to this weakness — prioritized sampling, learning from demonstrations, memory-efficient replay — are surveyed in §IV.B.

W3. Brittle exploration. ε-greedy action selection takes a random action with probability ε and a greedy action otherwise. This suffices for environments where reward is sufficiently dense that near-greedy policies explore the state space through their own exploitation. In environments where rewards are sparse or delayed beyond an ε-greedy random-walk's reach — Montezuma's Revenge [2], Pitfall! [2], Private Eye [2] — ε-greedy exploration is dithered rather than directed and fails to escape early-state plateaus. The formal characterization is that ε-greedy is myopic with respect to posterior uncertainty in $Q(s,a)$; methods responding to this weakness inject structured noise, maintain posterior estimates, or use intrinsic motivation, and are surveyed in §IV.C.

W4. Reward sparsity and credit assignment. When reward is received only at the end of a long trajectory, the one-step Bellman backup requires many iterations to propagate the signal to early states. Multi-step returns [19] partially mitigate this by allowing n-step bootstrapping:

$$y_t^{(n)} = r_t + \gamma r_{t+1} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n \max_{a'} Q(s_{t+n}, a').$$

But n-step returns trade variance for bias and do not address the deeper question of *what information* the return signal carries. Distributional RL [20] reframes the question: rather than estimating the expected return $\mathbb{E}[Z(s,a)]$, estimate the full distribution $Z(s,a)$. The distribution carries richer credit-assignment information — bimodality, skewness, tail behavior — that the scalar expectation discards. Methods responding to this weakness are surveyed in §IV.D.

W5. Distribution shift. Q-learning is off-policy *in principle*: the learned Q^* is independent of the behavior policy that generated transitions. In practice, the off-policy guarantee is fragile when the behavior policy is fixed and the support of the replay distribution is narrow — the regime of offline RL. The maximization $\max_{a'} Q(s',a')$ now ranges over actions a' that may be unsupported by the data, producing arbitrary extrapolation errors. The failure is structural: any algorithm that backs up through unsupported actions inherits unbounded error. Methods responding to this weakness constrain Q-estimates or actions to remain within the data support and are surveyed in §IV.E.

W6. Multi-agent coordination. When the environment contains multiple cooperating agents with a shared reward, naive per-agent Q-learning treats other agents as part of the environment — a non-stationarity that breaks the MDP assumption. Centralized Q-learning over the joint action space avoids this but scales exponentially: $|\mathcal{A}|^N$ for N agents. Value decomposition methods factor $Q_{\text{tot}}(\mathbf{s},\mathbf{a}) = f(Q_1(s_1,a_1),\ldots,Q_N(s_N,a_N))$ under structural constraints on f (additivity, monotonicity) that preserve the *individual-global-max* property — that the action maximizing Q_{tot} jointly is the per-agent argmax of each Q_i. Methods responding to this weakness are surveyed in §IV.F.

W7. Slow adaptation and sample throughput (composite axis). Vanilla Q-learning is bottlenecked along two distinct but interrelated dimensions that we treat as a single axis. The *adaptation* dimension: vanilla Q-learning trains a single Q-function for a single task; transfer to a related task — different reward, different transition dynamics, different state distribution — requires retraining from scratch or initialization from a pre-trained Q. Neither is satisfactory when adaptation must occur online and within a small number of episodes. The *throughput* dimension: a single sequential learner is bottlenecked on the rate at which the environment produces transitions, even with experience replay. Both dimensions admit the same set of methodological responses — distributed actor-learner architectures, recurrent value functions, meta-learning approaches, and the recent predictable-scaling literature — because at scale the distinction between "scale to adapt across tasks" and "scale to learn faster on one task" blurs (Agent57 is the paradigm case). We treat them as one composite axis rather than two separate ones because the methodological responses are mostly shared; readers who prefer a finer-grained taxonomy can read §IV.G as covering two sub-axes W7a (sample throughput / distributed systems) and W7b (slow adaptation / meta-learning), which we surface explicitly in §IV.G.A. Methods are surveyed in §IV.G.

W8. Function-approximation instability. The combination of off-policy learning, bootstrapping, and function approximation — the *deadly triad* [21] — is provably unstable in the general case. Concretely, the iterates of $Q(s,a;\theta) \leftarrow Q(s,a;\theta) + \alpha\big(r + \gamma \max_{a'} Q(s',a';\theta) - Q(s,a;\theta)\big)$ need not converge, may diverge, and even when they converge can do so to a point that is far from Q^*. Target networks, Polyak averaging, layer normalization, and architectural decomposition each address different aspects of this instability. Methods responding to this weakness are surveyed in §IV.H.

II.C. Notation conventions: Throughout the paper, s, a, r, s' denote a transition; θ and θ^- denote the parameters of an online and target network, respectively; π denotes a policy; α a learning rate; γ a discount factor; $\mathcal{D}$ a replay buffer; and ϵ either an exploration rate or a noise variable, disambiguated by context. Deviations from this convention are explicitly flagged in each section.

III. Methodology

This review surveys Q-learning algorithms spanning over three decades of development, from foundational tabular approaches to recent deep variants. We include influential papers that shaped the theoretical landscape, introduced practical improvements, or demonstrated wide applicability in real-world domains. The selection prioritizes methods whose contributions can be mapped onto one or more of the eight weakness axes introduced in §II.B — methods that respond to a recognizable weakness of vanilla Q-learning with a distinct mechanism.

A. Source selection and review protocol: This paper is best characterized as a *narrative review with empirical add-ons* rather than a strictly systematic review in the PRISMA sense: the synthesis is interpretive (each axis carries an argued reading of the methods it surveys), and the eight-axis framework was developed iteratively with the literature rather than applied as a pre-registered classification. We document the search and selection protocol here so that the resulting axis assignments, benchmark extractions, and repository comparisons can be independently audited.

Databases and search strategy. Source candidates were identified from three databases: Google Scholar (broad coverage, citation-graph traversal), arXiv (cs.LG and cs.AI listings, primary source for preprints), and Semantic Scholar (citation-relationship analysis). The search ran from December 2025 through May 2026 with a cutoff date of 2026-05-15 for the late-2025 / 2026 paper sweep documented separately. Searches combined topical anchors ("Q-learning", "deep Q-network", "DQN") with axis-specific terms ("overestimation bias", "prioritized replay", "exploration bonus", "distributional reinforcement learning", "offline reinforcement learning", "value decomposition", "distributed reinforcement learning", "meta-reinforcement learning", "function approximation stability") and recency filters (publication date 2018-onward for deep-RL methods; full historical range for foundational results).

Inclusion and exclusion criteria. Five inclusion criteria were applied:

1. **Method centrality** — the work introduces a methodological contribution to the Q-learning family rather than applying Q-learning to a domain.
2. **Mechanism distinctness** — the contribution is mechanistically distinguishable from prior work along at least one of the eight axes of §II.B.
3. **Empirical validation** — the work reports results on a recognized benchmark (Atari, classic control, D4RL, SMAC, ProcGen, or equivalent).
4. **Cross-reference impact** — the work is cited as foundational by subsequent methods in the same axis-family, *or* the work was published within the previous twelve months and represents an active research thread.
5. **Reproducibility** — code is publicly available or the algorithm is sufficiently specified to be independently re-implemented.

Exclusion criteria removed: (a) application-domain papers using Q-learning as a black-box tool without methodological contribution; (b) workshop-only or non-archival preprints lacking subsequent follow-up; (c) methods that have been demonstrably superseded by equivalent-effort alternatives (e.g., we cover Maximin Q-Learning but not all of its precursors). Methods satisfying all five inclusion criteria received per-paper treatment. Methods satisfying a subset received compact treatment as part of a family (e.g., the value-decomposition family in §IV.F is treated through its canonical members VDN, QMIX, QPLEX, QTRAN, with QFIX added on the basis of criterion 4's twelve-month clause).

Screening flow. Initial candidate identification produced approximately 200 papers; first-pass title/abstract screening narrowed this to approximately 120. A subsequent full-text screening against the five inclusion criteria produced approximately 80 papers receiving full per-method treatment, with an additional 20 cited at family or cross-reference level. We do not publish a strict PRISMA flow diagram here because the search-and-iterate process was non-linear: as the eight-axis framework crystallized, several methods that initially appeared incidental were promoted to per-paper status, and conversely some once-canonical entries were demoted to family-level mention. We treat this iteration as a methodological *limitation* rather than a defect (and surface it explicitly in §VIII.D); a strictly systematic review would have required pre-registering the axis framework before the literature search, which we did not.

Axis-assignment protocol. Each method covered in §IV is assigned to a *primary axis* — the weakness whose response motivated the method's introduction — and zero or more *secondary axes* — additional weaknesses the method incidentally addresses or partially mitigates. Primary assignment is made on the basis of the method's stated motivation in its original publication and the predominant mechanism of its contribution. Where the original publication's motivation differs from our axis assignment (notably for distributional methods in §IV.D and Dueling DQN in §IV.H), the re-interpretation is argued explicitly within the section. The complete per-method mapping, including secondary axes and cross-references, is provided as supplementary material.

Per-paper extraction template. For methods receiving per-paper treatment, we extracted: (i) the formal weakness statement the method addresses (§IV.X.A reference); (ii) the key mechanism- defining equation; (iii) the primary empirical evidence reported in the original publication; (iv) trade-offs explicitly discussed by the authors; (v) any subsequent ablations or follow-up evaluations relevant to the trade-off discussion. This extraction template is not published as supplementary material in the current version, but the per-method writeups in §IV reflect its structure consistently.

B. Comparison with prior surveys: To position this survey against existing work, we compiled Table I, which compares this paper with five prior Q-learning–focused surveys [7–10] and [11] along five distinguishing dimensions: analysis of public DQN repositories, unified taxonomy spanning tabular and deep Q-learning, extraction of original-paper Atari benchmarks, classic control benchmarks from controlled re-implementation, and per-paper review organized around mechanism and axis. Where prior surveys focus on chronological or method-type organization, this paper's problem-first axis structure (§IV) is, to our knowledge, the first multi-axis problem-first treatment of Q-learning specifically. This claim is supported by a documented 2024-2026 sweep across Google Scholar, arXiv (cs.LG and cs.AI), and Semantic Scholar that surveyed every recent Q-learning, DQN, and broad-RL survey we could identify; the closest prior art is the single-axis problem-first organization of [22], which addresses distribution shift in offline RL only. Search strings, dates, and per-survey notes are recorded as supplementary material so the claim can be independently audited.

C. Axis assignment methodology: Each method covered in this paper is assigned to a *primary axis* — the weakness whose response motivated the method's introduction — and zero or more *secondary axes* — additional weaknesses the method incidentally addresses or partially mitigates. Primary assignment is made on the basis of the method's stated motivation in its original publication and the predominant mechanism of its contribution. Where the original publication's motivation differs from our axis assignment (notably for distributional methods in §IV.D and Dueling DQN in §IV.H), the reinterpretation is argued explicitly within the section.

Cross-references between axis-sections — e.g., Rainbow appearing in §IV.B (primary) and being cross-referenced from §IV.A, §IV.C, §IV.D, and §IV.H — surface the connective tissue that catalogue-style surveys obscure. The method-type taxonomy table in §IV gives the category-level mapping; a full per-method index supporting readers who arrive expecting the older organization is provided as supplementary material.

D. Empirical evidence sources: Section V analyzes Atari benchmark results extracted from the original papers introducing each method. Section VI reports tabular benchmarks from controlled re-implementations on Gymnasium environments (FrozenLake-v1, Taxi-v3, CliffWalking-v1). Section VII analyzes algorithmic coverage across six widely-used open-source deep RL repositories (Tianshou [12], XuanCe [13], CleanRL [14], DQN Zoo [15], Stable Baselines3 [16], RLlib [17]). The three evidence streams together support different aspects of the paper's contribution: the literature analysis demonstrates methodological diversity, the controlled experiments isolate algorithmic from architectural effects, and the repository analysis maps the practical landscape of available implementations.

IV. Related Works

Over the past three decades, Q-learning and its deep variants have evolved through a long sequence of mechanism-level innovations. Section II.B identified eight foundational weaknesses of vanilla Q-learning that motivate the field's trajectory; this section surveys the methods that respond to each. The organization is problem-first: each subsection corresponds to one weakness, groups responses by mechanism, and compares them on the trade-offs they introduce.

The method-type taxonomy. Prior Q-learning surveys [7–10] organize methods by *type* — what kind of mechanism a method is. We call this the *method-type taxonomy* and use that term throughout. The table below lists its six categories and, crucially, shows where each category's methods fall among the eight problem axes. The categories cut *across* the

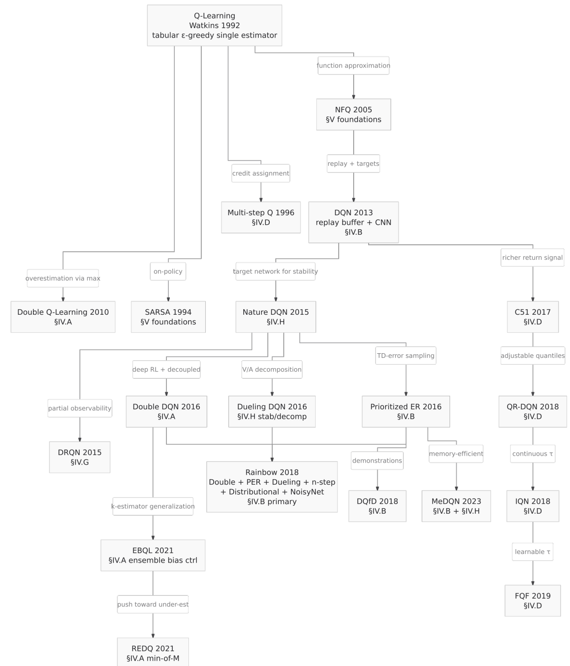

Fig. 1. Method genealogy of Q-learning and its deep variants. Edges are labeled by the weakness of the parent that the child method addresses.

axes: statistical methods split between exploration (§IV.C) and credit assignment (§IV.D), and ensemble methods between overestimation control (§IV.A) and exploration (§IV.C). A problem-first organization regroups exactly these cross-cutting cases by the weakness each method addresses rather than by what the method is. (The full per-method mapping is provided as supplementary material.)

This subsection orients the reader through three further artifacts: a method genealogy showing the inheritance relationships between Q-learning's variants, a companion figure showing the modern-RL branches that have emerged outside the conventional taxonomy, and an axis × mechanism-family matrix summarizing what mechanism types are deployed against each weakness. The §IV subsections that follow each provide a five-part structure: formal statement of the weakness (subsection A), grouped solution families (subsection B), trade-offs (subsection C), empirical evidence (subsection D), open questions (subsection E), and a comparison summary with positioning grid (subsection F).

A. Method genealogy: Figure 1 shows the genealogy of Q-learning methods covered in this paper. Nodes are methods; edges are inheritance relationships labeled by *the weakness of the parent that the child method addresses*. The annotations on the edges are themselves a contribution: they trace the field's evolution as a sequence of targeted responses rather than a chronological accumulation of techniques.

Two structural observations emerge from the genealogy. First, **Rainbow [6] is the integration point** for five separate axis responses (Double DQN, PER, Dueling, multi-step, distributional, NoisyNet) — it is not a single method but a deliberate composition across axes. Second, **the distributional family**

TABLE I
COMPARISON OF Q-LEARNING AND DEEP Q-LEARNING SURVEY PAPERS.
● = DISCUSSED; ○ = NOT DISCUSSED; PARTIAL = *PARTIAL*.

Aspect	Urtans 2018	Jang 2019	Boppiniti 2021	Hafiz 2022	Ghasemi 2024/25	Ours (2
Analyzes Public DQN Code Repositories	○	○	○	○	○	●
Unified Taxonomy Covering Both Tabular Q and DQN	○	○	○	○	*partial*	●
Atari Benchmark Results Extracted from Prior DQN Papers	○	○	○	○	○	●
Classic Control Benchmark Results from Original Implementations	○	○	○	○	○	●
Thorough Per-Paper Literature Review	○	○	○	○	*partial*	●
Problem-First Organization Across Multiple Axes	○	○	○	○	○	●

TABLE II
THE METHOD-TYPE TAXONOMY USED BY PRIOR Q-LEARNING SURVEYS — ITS SIX CATEGORIES AND WHERE EACH CATEGORY'S METHODS FALL AMONG THIS PAPER'S EIGHT PROBLEM AXES. FAMILIES THAT POSTDATE THIS TAXONOMY (OFFLINE RL §IV.E, MULTI-AGENT §IV.F, DISTRIBUTED/META §IV.G) HAVE NO METHOD-TYPE CATEGORY AT ALL.

Method-type category	Groups methods that…	Examples	Problem axes
Statistical	model the return distribution or inject parameter noise	C51, QR-DQN, NoisyNet	§IV.C, §IV.D
Q-Function Computation	alter how the bootstrap target is computed	Double DQN, Dueling, Rainbow	§IV.A, §IV.B, §IV.H
Memory / Replay	change what experience is stored and replayed	DQN, PER, DQfD, HER	§IV.B
Ensemble-Based	maintain several Q-estimators	Bootstrapped DQN, EBQL, REDQ	§IV.A, §IV.C
Model-Based	use a model or a posterior over values	Bayesian Q, Posterior-Sampling DQN	§IV.C, §V
Pure Q-Learning	tabular or minimal value iteration	Q-learning, SARSA, NFQ	§V, §IV.H

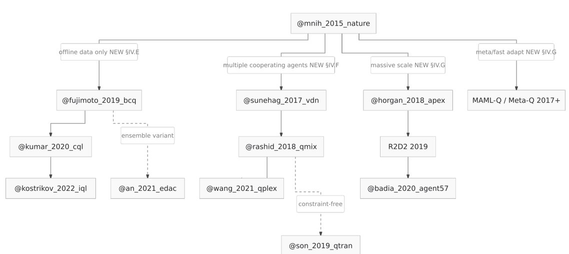

Fig. 2. Q-learning branches outside the conventional six-category mechanism taxonomy — offline RL (§IV.E), multi-agent value decomposition (§IV.F), and distributed / meta-learning (§IV.G).

(C51 → QR-DQN → IQN → FQF) constitutes an unusually long single-axis lineage compared to most other branches, reflecting the field's progressive refinement of distributional Q-learning under the credit-assignment weakness (W4).

B. Branches outside the conventional taxonomy: Figure 2 below shows three Q-learning branches that emerged outside the conventional six-category mechanism taxonomy — families that respond to weaknesses (distribution shift, multi-agent coordination, distributed scale and meta-adaptation) that the method-type taxonomy has no category to hold. These are the methods that motivate §IV.E, §IV.F, and §IV.G.

The visual absence of these branches from prior Q-learning surveys [7–10] is one of the most direct empirical arguments for the structural pivot of §IV. The method-type taxonomy classifies methods by mechanism type; these branches require new mechanism categories (constrained policies, value decomposition, distributed actors, meta-learning) that did not exist when the original taxonomy was conventionalized.

C. Axis × mechanism-family matrix: The matrix below summarizes which mechanism families address which axis. Filled cells indicate a primary contribution; open cells (○) indicate an incidental or secondary contribution; blanks indicate the axis is not addressed by that mechanism family.

Several patterns are visible at a glance:

- The **architectural decomposition** column is the most populous — most axes have at least one architectural response. This reflects the field's strong preference for *structural* solutions (modifying what the network represents) over *training-procedure* solutions (modifying what the loss optimizes).
- The **distributed** column is sparse but increasingly populated by frontier agents (Ape-X, Agent57). The pattern suggests scaling-as-mechanism is undercovered in the literature relative to its empirical importance — a point the open-questions subsection of §IV.G develops.
- The **decoupling** column has only two entries (DoubleQ-derived). Decoupling is mechanistically narrow; ensembling subsumes most of its effect once compute budgets permit $K > 2$ Q-functions.
- The **bottom-left quadrant** of the matrix (W6, W7 × decoupling or ensembles) is empty. Whether this reflects a true gap in the literature or merely a labeling artifact is an open question flagged in §VIII.

D. Section roadmap: The remainder of §IV proceeds axis by axis:

- **§IV.A** Overestimation bias — Double Q-learning to ensemble bias control (EBQL, REDQ).

Axis	Decoupling	Ensembles	Architectural decomposition	Behavior / loss modulation	Distributed / parallel
W1 Overestimation (§IV.A)	DoubleQ, DDQN	EBQL, REDQ	Dueling (rel.) ∘	—	—
W2 Sample inefficiency (§IV.B)	—	—	—	PER, DQfD, MeDQN, HER	Ape-X (replay) ∘
W3 Brittle exploration (§IV.C)	—	Bootstrapped, UCB-Q	NoisyNet, ParSpaceNoise	CBDQ, PSDQN, RND, Go-Explore	Agent57 (portfolio)
W4 Reward sparsity (§IV.D)	—	—	Distributional family	n-step, λ-returns	—
W5 Distribution shift (§IV.E)	BCQ	EDAC	—	CQL, IQL, BRAC, AWAC	—
W6 Multi-agent coord. (§IV.F)	—	—	VDN, QMIX, QPLEX, QTRAN	—	—
W7 Scaling / adaptation (§IV.G)	—	—	DRQN, R2D2	—	Ape-X, Agent57, Meta-Q
W8 Function-approx stability (§IV.H)	—	—	Target net, Dueling, PQN	MeDQN consol., Munchausen	—

- **§IV.B** Sample inefficiency — prioritized replay, demonstrations, memory-efficient consolidation, hindsight relabeling.
- **§IV.C** Brittle exploration — noise injection, ensemble disagreement, belief modulation, intrinsic motivation, archival exploration.
- **§IV.D** Reward sparsity and credit assignment — multi-step returns and the distributional family.
- **§IV.E** Distribution shift (offline RL) — policy constraint, value penalty, expectile regression, ensemble diversification.
- **§IV.F** Multi-agent coordination — additive, monotonic, duplex-dueling, and constraint-free value decomposition.
- **§IV.G** Scaling and slow adaptation — distributed actor-learner architectures, bandit-controlled exploration meta-policies, meta-learning, recurrence.
- **§IV.H** Function-approximation instability — target networks, architectural decomposition, normalization recipes, consolidation losses.
- **§IV.I** Theoretical foundations and recent advances — convergence theory, finite-time bounds, pessimism in offline RL, stability theory, and IGM theorems. A meta-section that surveys the theoretical landscape underlying the eight axes above.
- **§IV.J** Q-learning for foundation model alignment — Q-Transformer, ShiQ, VLM Q-Learning, $Q^\sharp$. An application-oriented section covering Q-learning's role in fine-tuning large language and vision-language models, distinct from the weakness-axes in that it represents a new deployment regime rather than a new mechanism category.

Each axis subsection (A–H) ends with **F. Comparison summary**: a uniform six-column table (method / mechanism category / mechanism / cost / best-at / empirical anchor) and, where the trade-off is naturally two-dimensional, a 2D positioning grid. §IV.E additionally provides a method-selection decision tree for practitioner use. §IV.I substitutes a results-by-year theoretical-status table for the comparison grid since its entries are theorems rather than methods. §IV.J provides a four-column comparison table organized by off-policy support and KL-regularization compatibility.

IV.A. Overestimation Bias

Weakness. The update target $r+\gamma\max_{a'}Q(s',a')$ applies a maximum over noisy estimates, which is biased upward: the expected maximum of noisy estimators exceeds the maximum of their true means [23]. Propagated through the Bellman backup the bias compounds, steering the greedy policy toward poorly-estimated actions rather than genuinely good ones — bounded in the tabular case but able to grow without bound under neural approximation.

Mechanisms. Three families debias the maximum, distinguished by *what information they exploit*. (i) *Decoupling* separates action selection from evaluation: tabular Double Q-learning [24] uses two independent estimators, and Double DQN [25] reuses DQN's target network for evaluation at near-zero cost, $y=r+\gamma Q(s',\arg\max_{a'}Q(s',a';\theta);\theta^-)$. A recent re-examination, DDQL [26], shows this target-network substitution is *not* equivalent to true two-estimator decoupling — it carries a stale-bootstrap bias — and training two independent networks restores lower bias and improves on 47 of 57 Atari games under matched compute. (ii) *Ensemble* methods generalize to K estimators and make the bias–variance trade-off tunable: EBQL [27] evaluates with the average of held-out members ($K\in[5,10]$ interpolating between DQN's over- and Double DQN's under-estimation), while REDQ [28] takes a minimum over a random subset to deliberately induce under-estimation. (iii) *Architectural* decomposition — Dueling DQN [29] — routes learning through a state-value baseline $V(s)$, reducing the relative noise in the action-conditional term; this is primarily a stability contribution ([§IV.H](#sec-iv-h)) that incidentally dampens the max bias. For *hybrid* discrete–continuous actions, PDQN [30] debiases the discrete max while a learned actor emits the continuous parameters, avoiding bias compounding across the two levels.

Trade-off. No fix dominates. Double DQN trades a small under-estimation for removing a large over-estimation at essentially no cost; ensembles buy *calibratable* bias at $K\times$ compute and memory; REDQ's minimum helps when the optimizer drifts toward high-Q regions but hurts when optimism is needed, as in long-horizon tasks that depend on exploration. The right operating point therefore depends on action-space geometry and on whether the downstream policy is more

TABLE III
OVERESTIMATION-BIAS METHODS (SEE SECTION V FOR BENCHMARK PROTOCOL CAVEATS).

Method	Mechanism	Cost	Best at
Double Q (2010)	Two swapped estimators	None	Tabular MDPs
Double DQN (2016)	Online selects, target evaluates	Free	Most Atari
DDQL (2026)	Two independent nets	2×	47/57 Atari
EBQL (2021)	Avg. of $K-1$ held-out	$K\times$	Bias calibration
REDQ (2021)	Min of M random members	$K\times$	Continuous control
Dueling* (2016)	$V(s)$ + centered $A(s,a)$	Modest	Many-action states

*Primary axis is stability (Section IV-H); listed here as an incidental contributor.

sensitive to over- or under-estimation. Critically, these *online* fixes transfer imperfectly to offline RL, where overestimation re-appears as out-of-distribution action selection and demands different tools (CQL, IQL; [§IV.E](#sec-iv-e)).

Open questions. The bias–variance Pareto frontier as a function of ensemble size K, subset size M, and update rate is unmapped; and the relationship between online debiasing and offline OOD-penalization remains ad hoc, lacking a framework that recovers both as limits of a single inequality.

IV.B. Sample Inefficiency

Weakness. Vanilla Q-learning consumes each transition once; experience replay [18] relaxes this by sampling minibatches from a buffer $\mathcal{D}$, $\theta \leftarrow \theta - \alpha\,\mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}[\nabla_\theta(Q(s,a;\theta)-y)^2]$. But uniform sampling treats every transition as equally informative, so when high-information transitions are rare — sparse rewards, low-probability states, narrow-margin decisions — learning stays bottlenecked on the rate at which those transitions reappear in the sample.

Mechanisms. Four families recur, distinguished by *what they manipulate*. (i) *Prioritized sampling*: Prioritized Experience Replay (PER) [31] draws transitions in proportion to TD-error magnitude, $P(i) \propto (|\delta_i|+\epsilon)^\alpha$, with importance-sampling weights annealed toward 1 to correct the induced bias; it is among the two largest contributors to Rainbow [6]. (ii) *Demonstration augmentation*: Deep Q-learning from Demonstrations (DQfD) [32] seeds the buffer with expert trajectories and adds a large-margin supervised term, trading exploration cost for demonstration cost to produce far stronger initial policies. (iii) *Memory-efficient consolidation*: MeDQN [33] replaces most of the buffer with a consolidation loss that distills past Q-values from the target network, cutting Atari storage roughly tenfold; the same mechanism damps function-approximation drift and so doubles as a stability tool ([§IV.H](#sec-iv-h)). (iv) *Goal relabeling*: Hindsight Experience Replay (HER) [34] rewrites failed goal-conditioned trajectories as successes under synthetic goals $g' = s_T$, densifying reward at no environmental cost.

Trade-off. No method dominates, because the relevant axes are categorical rather than a clean bias–variance frontier. PER's prioritization is asymptotically unbiased but injects variance early in training, tuned through α and the β schedule. DQfD's gain scales with demonstration quality and is inapplicable when expert data is absent or reward-misaligned. MeDQN trades buffer storage for the compute of its consolidation loss plus a weight λ to tune per domain. HER applies only to MDPs whose reward decomposes over goals, leaving non-goal-conditioned sparsity untouched. The choice therefore turns on environment structure and resource budget, not a single dominant operating point. Crucially, these gains are tied to the *online* replay distribution: PER's correction is computed against the current buffer, so its transfer to the fixed distribution of offline RL ([§IV.E](#sec-iv-e)) is unsettled and empirically mixed.

Open questions. FIFO replacement is the unjustified default and discards the rare early successes of highest information density, leaving adaptive, utility-aware replacement largely unexplored. And goal relabeling beyond Cartesian goals — rewards computable only from full trajectory features — lacks a parameterizable relabeling space.

IV.C. Brittle Exploration

Weakness. ε-greedy takes a uniform-random action with probability ε and otherwise acts greedily — *dithered* noise applied independently per step, with no memory of past exploration and no model of which actions remain uncertain. It is myopic with respect to epistemic uncertainty: a state-action pair visited many times is sampled at the same rate as one visited never, so it fails when reward is sparse, delayed, or behind passages whose random-walk hitting time is exponential in trajectory length — the canonical case being Montezuma's Revenge [2], where DQN scores essentially zero.

Mechanisms. Four families induce a posterior over $Q(s,a)$ or an exploration bonus, distinguished by *what they exploit*. (i) *Noise injection* perturbs the network rather than the action: parameter-space noise [35] adds $\mathcal{N}(0,\sigma^2 I)$ to θ once per episode (with layer normalization [36] and adaptive σ) for temporally consistent exploration, while NoisyNet [37] replaces weights with learned $\mu + \sigma \odot \varepsilon$ sampled per forward pass so the network *learns* where to keep exploring — a load-bearing component of Rainbow [6]. (ii) *Ensemble disagreement* treats variance across K Q-heads as epistemic signal: Bootstrapped DQN [38] samples one bootstrap-trained head per episode, and UCB Q-Ensembles [39] act on $\arg\max_a(\mu(s,a)+\lambda\sigma(s,a))$ — the same ensemble that serves bias control in [§IV.A](#sec-iv-a). (iii) *Belief / posterior modulation* reweights backups: CBDQ [40] maintains a clustered belief $b_t(a \mid s)$ that weights future values, while Posterior Sampling DQN [41] draws a Q-hypothesis from an approximate posterior [42] per episode (deep Thompson sampling). (iv) *Intrinsic motivation* modifies reward: RND [43] adds a bonus from random-network prediction error that decays as states become familiar, the first method to reach non-trivial Montezuma scores; Go-Explore [44] instead archives visited states and *returns-then-explores*, decoupling reaching a state from exploring it, and fully solves Montezuma's Revenge.

Trade-off. No method dominates. Per-episode perturbation (parameter noise, Bootstrapped DQN) yields multi-step consis-

TABLE IV
SAMPLE-EFFICIENCY METHODS.

Method (year)	Mechanism	Cost	Best at
DQN replay (2013)	Uniform sampling from buffer	Memory	Off-policy baseline
PER (2016)	Sample $\propto \lvert\delta_i\rvert^{\alpha}$, IS-corrected	IS bias, α tuning	Rare high-info transitions
DQfD (2018)	Buffer seeded with expert demos	Demonstration availability	Sparse-reward Atari
MeDQN (2023)	Consolidation loss compresses buffer	λ hyperparameter	Memory-constrained training
HER (2017)	Goal relabeling for synthetic reward	Goal-conditioned only	Robotic manipulation

tency that solves partially-observed tasks like flickering Pong [45], whereas NoisyNet's per-pass sampling adapts faster but loses that property. Ensembles multiply forward-pass cost and memory by K (largely hidden by batching). Intrinsic bonuses add a scale hyperparameter and risk novelty-addiction that crowds out extrinsic reward. Go-Explore is empirically dominant on hard-exploration Atari but introduces a non-Markovian archive whose extension to continuous spaces is unsettled. Crucially, demonstrations remain the most effective sidestep on the hardest games — Rainbow's best honest Montezuma result trails DQfD by an order of magnitude — sharpening the exploration-vs-demonstration tension of [§IV.B](#sec-iv-b); distributional gains ([§IV.D](#sec-iv-d)) do not transfer to this axis at all.

Open questions. All methods here fix one exploration strategy for all of training, yet game-by-game variance argues for adaptive selection (meta-learned, bandit-controlled as in Agent57 [46] [§IV.G](#sec-iv-g), or uncertainty-thresholded), and no consensus exists on balancing intrinsic against extrinsic reward or on archive-based exploration in continuous latent spaces.

IV.D. Reward Sparsity and Credit Assignment

Weakness. The one-step Q-learning update propagates reward by exactly one Bellman backup per environment step, so when reward arrives only at the end of a long trajectory the signal must traverse the whole horizon through repeated updates before it reaches early states. This is the credit-assignment problem: apportioning fractional responsibility for a terminal reward across upstream state-action pairs, which a scalar expected-return target conveys slowly and with little structure.

Mechanisms. Two families attack the weakness by changing *what the return signal carries*. (i) *Temporal-extent* methods widen the backup: Multi-Step Q-Learning [19] uses n-step bootstrapped returns with eligibility traces, $y_t^{(n)} = r_t + \gamma r_{t+1} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n \max_{a'} Q(s_{t+n}, a')$, and the TD($\lambda$) interpolation tunes bias against variance via λ; in Rainbow [6] three-step returns are the second-largest performance contributor after prioritized replay. (ii) *Information-content* methods replace the scalar $\mathbb{E}[\text{return}]$ with the full return distribution $Z(s,a)$. C51 [20] models Z as a categorical distribution over 51 fixed atoms on a bounded support $[V_{\min}, V_{\max}]$ and minimizes a projected-KL target; QR-DQN [47] inverts this with fixed probabilities and *learned* quantile locations under a quantile Huber loss, removing the projection and the support prior; IQN [48] samples quantile fractions $\tau \sim \mathcal{U}(0,1)$ at runtime to learn a continuous distribution, reaching Rainbow-comparable scores alone; and FQF [49] additionally *learns* the fractions per state-action pair, attaining the highest median Atari at roughly 20% more compute than IQN.

Trade-off. No method dominates. Larger n or λ propagates reward faster but inflates variance and off-policy bias, so three-step is the practical sweet spot for single-stream agents while distributed throughput (Ape-X, R2D2) makes longer horizons viable. Across the distributional family, resolution trades against compute along the monotone-in-median diagonal C51 → QR-DQN → IQN → FQF, yet per-game behavior is non-monotone. The quantile-regression methods also incur a *quantile-crossing* pathology: training independent estimators per τ_i without enforcing $\tau_i < \tau_j \Rightarrow \theta_i \leq \theta_j$ yields non-monotone pseudo-distributions, worst early in training and on asymmetric rewards; non-crossing variants smooth the curves at small median gains. Interpretively, this paper reframes the distributional family as a *credit-assignment* mechanism rather than the conventional uncertainty grouping ([§IV.G](#sec-iv-g)): all these methods still *act* on $\mathbb{E}[Z(s,a)]$, so the richer distribution helps *learning*, not *acting*, and its strength on long-horizon tasks reflects better backup propagation, not optimism over uncertainty. The flip side is that richer signal cannot substitute for directed search — on the hardest exploration games ([§IV.C](#sec-iv-c)) the family scores near zero, sharpening the W3/W4 separation.

Open questions. Convergence guarantees for the quantile family under function approximation remain substantially weaker than for C51's projection step, and whether its compositional gain with prioritized replay is orthogonal or partly redundant is unsettled given IQN's standalone strength. Whether these architectures support stable risk-sensitive action selection on tail behavior, rather than acting on the mean, is largely unexplored.

IV.E. Distribution Shift (Offline Reinforcement Learning)

Weakness. Q-learning's off-policy property is theoretical: the Bellman operator is independent of the data-generating policy, but in *offline* RL the buffer is fixed and the bootstrap target $r + \gamma \max_{a'} Q(s', a')$ ranges over the entire action space, including out-of-distribution (OOD) actions no behavior policy π_b ever took. The approximator extrapolates arbitrarily at such points, and iterated backups drive Q unboundedly upward where data support vanishes — so naive offline Q-learning consistently underperforms even the behavior policy [50].

Mechanisms. Four families respond, distinguished by *what they constrain*. (i) *Policy constraint* restricts the learned pol-

TABLE V
EXPLORATION METHODS.

Method (year)	Mechanism	Cost	Best at
Param Space Noise (2017)	Per-episode θ perturbation	Layer norm	Episode-consistent exploration
NoisyNet (2018)	Per-pass weight noise, learned σ	2× forward	Adaptive exploration
Bootstrapped DQN (2016)	K-head ensemble, head per episode	$K\times$ memory	Episode-coherent exploration
UCB Q-Ensemble (2018)	$\mu + \lambda\sigma$ over K-ensemble	$K\times$ compute	Uncertainty-aware action
CBDQ (2025)	Belief over actions weights backups	Clustering	Classic control + driving
Posterior Sampling DQN (2023)	Sample Q from posterior per episode	Approx. posterior	Cyclic environments
RND (2018)	Random-network distillation bonus	Bonus scale	Sparse-reward Atari
Go-Explore (2019/21)	Archive + return-then-explore	State hashing	Hardest Atari exploration

TABLE VI
CREDIT-ASSIGNMENT AND DISTRIBUTIONAL METHODS.

Method (year)	Mechanism	Cost	Best at
Multi-Step Q (1996)	n-step returns + eligibility traces	Variance under off-policy	Long-horizon credit assignment
C51 (2017)	51 fixed atoms, projected KL	Bounded support pre-specified	Atari mean + median
QR-DQN (2018)	N learned quantile locations	Quantile loss, no projection	Dense + strategic Atari
IQN (2018)	Runtime-sampled $\tau \sim U(0, 1)$	Cosine fraction embedding	Standalone Rainbow-comparable
FQF (2019)	Learned quantile fractions + values	+20% compute vs. IQN	Highest median Atari

icy toward π_b: BCQ [50] samples actions from a learned behavior model plus small perturbations; BRAC [51] adds an explicit KL/Wasserstein divergence penalty; and AWAC [52] uses an advantage-weighted update $\pi(a \mid s) \propto \pi_b(a \mid s)\exp(A(s, a)/\beta)$ that bridges to online fine-tuning. (ii) *Value penalty* — CQL [53] adds a regularizer that pushes Q down at OOD actions and up on in-distribution ones, yielding a lower bound on V^π. (iii) *Avoiding the max* — IQL [**?**] drops the max entirely, learning a state value via expectile regression and backing up $Q(s, a) \leftarrow r + \gamma V(s')$, so extrapolation error is structurally prevented. (iv) *Ensemble diversification* — EDAC [54] takes a min over K critics trained to disagree on OOD actions; Cal-QL [55] calibrates CQL's bounds for the offline-to-online bridge. A newer line couples expressive generative policies to Q-learning: FQL [56] trains a one-step flow-matching policy, sidestepping backpropagation through a diffusion chain, while adjoint-matching variants tolerate full multi-step generation and recover the optimal behavior-regularized policy at convergence.

Trade-off. No method dominates, and the axis trades cleanly against itself. Policy-constraint methods are safe but cannot outperform the best in-support trajectory by much; CQL's penalty weight α is its single most sensitive hyperparameter and varies across tasks despite adaptive scaling [57]; IQL avoids extrapolation but weakens formal optimality guarantees; EDAC's K-network ensemble multiplies compute and needs per-action gradients; flow policies trade expressiveness against tractability. The deeper connection is to [§IV.A](#sec-iv-a): offline OOD action selection *is* overestimation bias under a fixed buffer — the online max-over-noisy-estimates problem re-appearing where the correction signal of fresh on-policy samples is absent. The online debiasing tools transfer imperfectly, which is precisely why the offline setting demands its own conservatism (CQL, IQL) rather than the decoupling and ensembles that suffice online.

Open questions. A unifying framework recovering all four families as limits of a single regularization remains absent, and whether Q-learning's mechanism suits internet-scale heterogeneous data — where sequence-modeling approaches currently dominate — is empirically open with substantial practical stakes.

IV.F. Multi-Agent Coordination

Weakness. Cooperative multi-agent RL models N agents in a shared-reward dec-POMDP where each agent acts on a local observation-action history τ_i but the reward depends on the joint action. Independent Q-learning treats co-learners as non-stationary dynamics and destabilizes, while a centralized joint Q_{tot} over the product action space scales exponentially in N. The crux is the **Individual-Global-Max (IGM)** property: a factorization $Q_{\text{tot}} = f(Q_1, \ldots, Q_N)$ must let each agent act greedily on its local Q_i yet recover the joint greedy action, $\arg\max_{\mathbf{a}} Q_{\text{tot}} = \left(\arg\max_{a_1} Q_1, \ldots, \arg\max_{a_N} Q_N\right)$, so that decentralized execution stays globally optimal.

Mechanisms. Decomposition families trade representational capacity against training stability by how strictly they constrain f. (i) *Additive* — VDN [58] sets $Q_{\text{tot}} = \sum_i Q_i(\tau_i, a_i)$, trivially IGM-satisfying and trivial to train, but unable to express non-separable coordination. (ii) *Monotonic* — QMIX [59] relaxes additivity to $\partial Q_{\text{tot}}/\partial Q_i \geq 0$ via a state-conditioned hypernetwork mixer, preserving IGM with richer capacity and becoming the dominant baseline. (iii) *Full-expressiveness via dueling* — QPLEX [60] mixes per-agent advantage streams under an IGM-preserving constraint, representing the entire IGM class. (iv) *Constraint-free* — QTRAN [61] learns an unconstrained joint Q and enforces IGM through auxiliary losses, maximal in principle but hard to optimize.

Trade-off. The progression VDN → QMIX → QPLEX → QTRAN raises representational capacity monotonically, but capacity and stability are *not* independent: the empirical SMAC ordering is QMIX ≈ QPLEX > VDN > QTRAN,

TABLE VII
OFFLINE Q-LEARNING METHODS.

Method (year)	Mechanism	Cost	Best at
BCQ (2019)	Generative behavior model + perturbation	Generator quality	Narrow-support data
BRAC (2019)	KL/Wasserstein divergence penalty	Divergence weight tuning	Diverse offline data
AWAC (2020)	Advantage-weighted offline → online	β tuning	Offline-to-online bridge
CQL (2020)	Penalize Q at OOD actions	α sensitivity	Random/medium data
IQL (2021)	Expectile V, Q backed up via $V(s')$	τ tuning	Single-network simplicity
EDAC (2021)	Min over K-ensemble + gradient diversity	$K\times$ compute	Locomotion strongest
Cal-QL (2023)	Calibrated conservative bounds	α tuning	Offline-to-online bridge
FQL (2025)	One-step flow-matching policy + Q	Flow training	Expressive offline policies

because structural constraints that buy expressiveness also degrade training dynamics. QPLEX's full expressiveness pays off only on super-hard scenarios where representation is the bottleneck, whereas QTRAN's theoretical maximum never realizes — its loss-enforced IGM is the canonical case of capacity without trainability. QFIX/Q+FIX [62] reframes this tension as a *base-plus-correction* pattern rather than a monolithic mixer redesign: a small per-agent residual network corrects a VDN, QMIX, or QPLEX base toward the full IGM class, recovering the QPLEX expressiveness ceiling with simpler training dynamics and consistent gains over both QMIX and QPLEX on SMACv2 and Overcooked. This decouples the two axes that the earlier methods conflated. These instabilities compound the single-agent biases of [§IV.A](#sec-iv-a) and [§IV.H](#sec-iv-h), and the required centralized critic limits the framework to settings where centralized training is feasible.

Open questions. The IGM frame assumes a single optimal joint policy and greedy decentralized execution, leaving equilibrium selection, communication, and heterogeneous agents to ad-hoc extensions. Multi-agent off-policy correction [63] and the offline multi-agent regime — where per-agent behavior constraints need not yield coordinated joint behavior — remain among the cleanest cross-axis open problems.

IV.G. Scaling and Slow Adaptation

Weakness. This axis is *composite*: vanilla Q-learning is both sample-throughput-bound (W7a) and slow to adapt across tasks (W7b). A single sequential learner cannot consume transitions faster than the environment produces them — collecting Rainbow's [6] 200M Atari frames takes weeks of wall-clock time — while a Q-function trained on task $\mathcal{T}_1$ specializes to $\mathcal{T}_1$ and neither retraining nor warm-starting from $Q^{\mathcal{T}_1}$ yields few-shot transfer to a related $\mathcal{T}_2$. The two failure modes are bundled because the methods that answer either mostly answer both.

Mechanisms. The W7a (throughput) cluster parallelizes interaction. Ape-X [64] decouples collection from learning — many actors feed a centralized prioritized buffer that one GPU learner drains — for $\sim 50\times$ wall-clock speedup; R2D2 [65] adds an LSTM agent with replay *burn-in* to re-sync stale recurrent states, building on DRQN's [45] recurrent head for partial observability; Agent57 [46] adds an NGU intrinsic-motivation module and a bandit over an exploration-discount policy portfolio, the first to clear human level on all 57 games. PQN [66] instead uses synchronous vectorized environments, trading actor-learner staleness for single-machine simplicity ([§IV.H](#sec-iv-h)). The W7b (adaptation) cluster meta-trains across a task distribution $p(\mathcal{T})$, differing in *what is meta-learned*: an initialization (MAML-Q [67, 68], the first-order Reptile-Q [69], proximal ProMP [70])

$$\theta_0^* = \arg\min_{\theta_0} \mathbb{E}_{\mathcal{T}}\big[\mathcal{L}_{\mathcal{T}}(\theta_0 - \alpha\nabla_{\theta_0}\mathcal{L}_{\mathcal{T}}(\theta_0))\big];$$

a context embedding z that conditions $Q(s, a, z)$ with no gradient adaptation (PEARL [71], off-policy MQL [72]); or a forward-pass update rule, where a transformer reads the recent trajectory and predicts actions in-context — algorithm distillation and AdA, consolidated since 2023 into in-context Q-learning proper (SICQL [73], compositional ICQL [74]).

Trade-off. The deep distinction is between *hardware-driven* throughput and *algorithmic* few-shot adaptation. W7a methods buy performance with parallelism and compute: the Ape-X→R2D2→Agent57 progression is the single largest source of Atari gains since DQN, suggesting throughput is the binding constraint at scale and that algorithmic axes become secondary once compute is abundant. W7b methods instead buy data-efficiency on new tasks — adapting in 5–50 episodes where single-task baselines need thousands — at 10–100× meta-training cost whose payoff is empirically mixed. Methods span both because the distributed compute W7a developed is precisely what makes meta-training and large in-context models feasible, and because *implicit* meta-learning (a single Agent57/AdA-style agent trained across the whole distribution) often matches *explicit* outer-loop meta-learners while also exploiting the distributed actor pool. Predictable scaling laws [75] reframe both: the data-versus-compute Pareto follows the updates-to-data ratio, so the optimal allocation can be extrapolated from a small pilot — orthogonal to, and compatible with, the architecture that supplies the throughput.

Open questions. Whether meta-Q methods extend beyond narrow parametric task distributions to heterogeneous state/action/reward families is unresolved, and no compute-matched, axis-stratified ablation exists to separate genuine algorithmic gains from gains that are merely compensating for scale.

TABLE VIII
VALUE-DECOMPOSITION METHODS.

Method (year)	Representational capacity	Training stability	Core mechanism
VDN (2018)	Lowest (additive only)	Highest	$Q_{\text{tot}} = \sum_i Q_i$
QMIX (2018)	Medium (monotonic)	High	State-conditioned monotonic hypernetwork mixer
QPLEX (2020)	High (full IGM class)	Medium	Duplex dueling on advantage streams
QTRAN (2019)	Maximum (unconstrained)	Lowest	Unconstrained joint Q + auxiliary IGM losses
QFIX/Q+FIX (2025)	High (recovers IGM ceiling)	High	Per-agent residual correction on a base mixer

TABLE IX
DISTRIBUTED-SCALING AND ADAPTATION METHODS.

Method (year)	Sub-axis (W7a/W7b)	Mechanism	Best at
DRQN (2015)	W7b	LSTM head over DQN	Partial observability
Ape-X (2018)	W7a	Async actors + central learner + PER	50x wall-clock speedup
R2D2 (2019)	W7a	Ape-X + LSTM + replay burn-in	Memory-demanding tasks
Agent57 (2020)	W7a/W7b	R2D2 + NGU + bandit policy portfolio	All 57 Atari at human level
PQN (2024)	W7a	Synchronous vectorized envs + LayerNorm	Compute-efficient single machine
MAML-Q / Reptile-Q / ProMP (2017+)	W7b	Meta-learn fast-adapt initialization	Few-shot gradient transfer
PEARL / MQL (2019/20)	W7b	Inferred task-context embedding	Adaptation without gradient steps
SICQL / ICQL (2026)	W7b	In-context Q via transformer forward pass	Compositional task transfer

IV.H. Function-Approximation Instability

Weakness. The *deadly triad* [21] names the conjunction of off-policy learning, bootstrapping, and function approximation: any two are stable together, but all three admit counterexamples in which the semi-gradient update

$$\theta \leftarrow \theta + \alpha\big(r + \gamma \max_{a'} Q(s', a'; \theta) - Q(s, a; \theta)\big)\nabla_\theta Q(s, a; \theta)$$

diverges or oscillates. The instability arises because each gradient step also moves Q at the bootstrap target $Q(s', a')$, making the update self-referential and potentially expansive — deep Q-learning trained naively does diverge.

Mechanisms. Four families damp the triad, distinguished by *what they exploit*. (i) *Target decoupling* freezes a copy $Q(\cdot; \theta^-)$ for a window so that $y = r + \gamma \max_{a'} Q(s', a'; \theta^-)$ no longer chases its own parameters [1]; this is the single load-bearing stabilizer of the classical recipe, whether updated by hard copy or Polyak averaging. (ii) *Architectural decomposition* — Dueling [29] — routes most of the gradient through a state-value baseline, reducing noise in the action-conditional term; we re-interpret its contribution as primarily stability, consistent with the Rainbow ablation showing it the smallest-impact component [6] (its incidental bias effect appears in [§IV.A](#sec-iv-a)). (iii) *Normalization* damps the expansive map directly: PQN [66] applies LayerNorm [36] throughout the body, input BatchNorm, optional L2, and n-step returns, while Munchausen DQN adds a $\log \pi$ reward bonus that implicitly regularizes toward the previous iterate. (iv) *Consolidation* — MeDQN [33], kin to elastic weight consolidation [76] — penalizes movement away from past Q-values to counter the non-stationarity that drives drift.

Trade-off. No method dominates, because the recipes trade staleness, sample efficiency, and complexity against one another. Hard target updates buy stability at the cost of targets up to C steps stale; soft updates smooth that staleness but slow responsiveness; consolidation and Munchausen bonuses add a hyperparameter (λ, τ) that under- or over-damps. The pivotal result is PQN's: target networks *and* experience replay — long held essential — can be removed entirely, with LayerNorm plus parallel vectorized environments matching PER [31], Double DQN [25], and Rainbow [6] on Atari at up to $50\times$ less wall-clock. The triad's instability is thus dampable by normalization alone, the classical stabilizers being sufficient but not necessary. The catch is that PQN substitutes parallel environments for replay's sample reuse, so where interaction is expensive (real robotics) the trade-off may still favor replay retention.

Open questions. A formal stability guarantee for normalized Q-learning under the deadly triad — beyond informal Lipschitz arguments — remains absent, as does a direct test isolating Dueling's decomposition on vanilla DQN. Whether online normalization recipes transfer to offline regimes ([§IV.E](#sec-iv-e)), where extrapolation to unsupported actions is not damped by target networks, is largely unexplored.

IV.I. Theoretical Foundations and Recent Advances

Classical theory characterizes Q-learning in regimes today's methods have outgrown. Tabular Q-learning converges with probability one to Q^* under standard stochastic-approximation conditions [Watkins & Dayan 1992], the proof reducing the

TABLE X
STABILITY METHODS.

Method (year)	Mechanism	Cost	Best at
Target Network (2015)	Frozen Q copy for bootstrap target	Target staleness (C steps)	Foundational stability
Dueling DQN (2016)*	$V(s)$ + centered $A(s,a)$ decomposition	Modest	Many-action states
Munchausen DQN (2020)	$\log\pi$ reward bonus, implicit KL	τ hyperparameter	Implicit regularization
MeDQN (2023)	Past-Q consolidation loss	λ hyperparameter	Catastrophic forgetting
PQN (2024)	LayerNorm + n-step + parallel envs; no target net	Compute-structure shift	Modern norm-only recipe

Bellman optimality operator to an ℓ_∞ contraction. Once off-policy sampling, bootstrapping, and function approximation combine — the deadly triad of [§IV.H](#sec-iv-h) — iterates can diverge even when an optimal linear fit exists [Baird 1995; Tsitsiklis & Van Roy 1997], and no fixed-point theorem covers the general combined operator; the GTD/TDC family recovers off-policy linear convergence by minimizing an explicit MSPBE objective [77].

Recent work targets the neural and offline regimes where modern methods actually run. Offline RL is the most mature axis: pessimistic value iteration attains $\tilde{O}(\sqrt{1/n})$ suboptimality without behavior-coverage assumptions [Jin, Yang & Wang 2021], grounding the CQL/IQL penalties of [§IV.E](#sec-iv-e), with extensions to multi-task settings [78]. Finite-time bounds for neural fitted-Q hold only under restrictive complexity and mixing assumptions [79]. Stability theory is catching up to practice: capacity loss [80] and the broader plasticity-loss literature [Klein et al. 2026] now formalize why deep Q-networks degrade. Distributional methods enjoy clean tabular accounts — the distributional Bellman operator is a γ-contraction in Wasserstein-p [20], and quantile regression in Wasserstein-∞ [81] — but neural extensions remain open, as does a complete IGM characterization for multi-agent decomposition [Wang et al. 2020]. The recurring pattern is that empirical advance precedes theory by years.

Formal statements and proof sketches are provided in the supplementary material.

IV.J. Q-Learning for Foundation Model Alignment

Aligning large language and vision-language models recasts a familiar problem in unfamiliar dimensions, and that change of scale is what turns it into a Q-learning problem. Each generated token is an action drawn from a vocabulary of 10^4–10^5 entries, so a single response is a long trajectory through an enormous discrete action space. The reward is sequence-level and sparse: an entire generation receives one scalar from a learned reward model, with no built-in credit assignment to the tokens that earned it — a sparse-reward problem ([§IV.D](#sec-iv-d)) at extreme scale. Finally, alignment must not destroy the pretrained model's fluency, so the objective is regularized by a KL penalty against a reference policy, replacing the usual return with a return-minus-KL target:

$$Q_{\text{reg}}(s,a) = r(s,a) + \gamma\,\mathbb{E}\big[V_{\text{reg}}(s')\big] - \beta \log \pi_{\text{ref}}(a \mid s).$$

Methods cluster by how they exploit this structure. Offline trajectory and value methods reuse fixed data, including failed attempts: Q-Transformer [82] trains an autoregressive transformer Q-network over discretized action tokens with conservative penalties ([§IV.E](#sec-iv-e)), and VLM Q-Learning carries the same off-policy recipe to multimodal (image, text, action) data. A second group reinterprets the model's own machinery: ShiQ treats per-token logits as Q-values up to a bias correction, making temporal-difference learning directly compatible with the KL-regularized objective and removing the online rollouts that policy-gradient RLHF requires. A third, Q-sharp (Q), targets provable behavior by guiding the reference policy with a *distributional* Q-function, trading lower KL for a given reward gain. In-context variants — SICQL and ICQL — fit transformer Q-functions that adapt at inference time with no gradient updates, and reward-shaping approaches such as Q-shaping inject value estimates as auxiliary shaping signal rather than as the optimization target.

This is frontier, fast-moving work: the benchmarks are immature (reward models judging reward models), the methods are months apart, and value-based alignment may eventually fold into a broader axis on reward-model robustness. Its long-term place in the taxonomy is not yet settled, but its present importance to the field is.

V. ATARI BENCHMARK ANALYSIS

This section synthesizes Atari benchmark performance reported across the methods of §IV. Because third-party implementations often diverge from authors' codebases, and because official repositories are incomplete (§VII), we extract results directly from peer-reviewed publications rather than re-running experiments. Tables II and III present raw per-game scores under each paper's original evaluation protocol; Figure F-B2 (if included) provides an axis-stratified visual summary.

A. Table structure and dual-view organization: Tables II and III are structured on two axes simultaneously, and both groupings are load-bearing.

Row grouping (the conventional six-category taxonomy). Methods are grouped into six rows: *Statistical Methods*, *Q-Function Computation*, *Memory/Replay*, *Ensemble-Based*, *Model-Based*, and *Pure Q-Learning*. This grouping matches the method-type taxonomy used by prior Q-learning surveys [7–10]. We retain it in Tables II/III as a presentation device, because (a) it lets readers familiar with the conventional taxonomy navigate the empirical data without translation; (b) it preserves within-family comparability across the historical literature; and (c) the six-category grouping makes the modern-RL families (§IV.E offline, §IV.F multi-agent, §IV.G distributed) visually salient as absences — there is no six-category row for them to occupy.

Column grouping (task categories). Atari games are grouped into seven task categories — Reaction-Time Control, Strategic Planning, Sparse Rewards, Dense Rewards, Large Observation Space, Partially Observable Environments, Stochastic Environments.

Axis annotation. A rightmost column or footnote on each table maps each method's row to its primary §IV axis. The annotation does not change the table's structure; it provides a third lens. Readers can read Table II as "method-family × task-category" by attending to the row grouping, or as "primary-axis × task-category" by attending to the axis column.

The dual-view organization is itself a contribution. Readers arriving with the catalogue-style expectation find the familiar six-category structure; readers seeking analytical synthesis follow the prose subsection-by-subsection below and the axis column. Neither audience needs to translate the empirical data to align with their reading.

B. Tables II and III — extracted scores:

C. Task category stratification: **Each Atari task category preferentially tests one or two of the eight weaknesses introduced in §II.B**:

The mapping is not perfect — most games test multiple weaknesses to some degree — but the dominant signal in each category aligns cleanly with one or two axes. The empirical claim of this paper's structural pivot is that methods targeting weakness W_i should excel on the diagnostic category for W_i and remain at baseline elsewhere. Tables II and III, read through this lens, support the claim.

D. Patterns by axis: **Brittle exploration (W3, §IV.C).** The sparse-reward category (Montezuma's Revenge, Pitfall!, Private Eye) is where the axis matters most. Methods targeting exploration explicitly — NoisyNet, Bootstrapped DQN, CBDQ, Posterior Sampling DQN — produce nonzero scores on Montezuma where vanilla DQN and methods targeting other axes do not. Rainbow's 384 on Montezuma is attributable specifically to its NoisyNet component (per the original ablation). DQfD's 4,638 on Montezuma — substantially higher than any non-demonstration method — illustrates the trade-off explored in §IV.B: demonstration substitutes for directed exploration. RND and Go-Explore (newly discussed in §IV.C but not in Tables II/III) report Montezuma scores of ~10,000 and >1 million respectively, indicating that the axis is not saturated by the methods covered here.

Reward sparsity and credit assignment (W4, §IV.D). The strategic-planning category (Q*bert, Ms. Pac-Man) is where distributional methods produce their most striking results. QR-DQN's 572,510 on Q*bert is the single largest score in our compilation by any method — over 25× the best non-distributional result on the same game — and the pattern is consistent across the category. IQN and FQF follow the same pattern at lower absolute magnitude.

Function-approximation stability (W8, §IV.H). Reaction-Time Control games (Breakout, Space Invaders, Enduro) show comparatively small inter-method variance: Double DQN, Dueling DQN, Rainbow, and distributional methods cluster within a factor of two of each other on these games. The pattern indicates that for games where the weakness signal is *stability rather than capability*, modest algorithmic improvements over the DQN baseline suffice — and PQN's strong performance on the category (§IV.H) is consistent with this interpretation.

Sample inefficiency (W2, §IV.B). PER's performance pattern, across the categories, is most pronounced on strategic-planning and sparse-reward games — categories where high-information transitions are rare. On dense-reward games where informative transitions are abundant under uniform sampling, the PER advantage is small. The ablation pattern from Rainbow (PER is the largest single contributor) is consistent with this category-stratified picture.

E. The reporting gaps as evidence: Dashes ("—") in Tables II and III indicate that the original paper introducing the method did not report a score for that game under the stated evaluation protocol. Absence of data does not imply poor performance — but the gap *pattern* is itself diagnostic and carries more information than is sometimes assumed.

For most methods, the gap pattern is itself diagnostic. A method that reports comprehensively on dense-reward games and skips sparse- reward games signals — perhaps implicitly — that the method does not solve the sparse-reward problem. A method that reports on Reaction- Time Control and skips Stochastic Environments is unlikely to be a strong response to W1 (overestimation, which compounds under stochasticity). The pattern of dashes, considered as a structured absence, is sometimes more informative than the present scores.

Several specific gaps illustrate the point. Ensemble Bootstrapping [27] reports on 11 of 57 games and skips the entire sparse-reward category, despite the ensemble mechanism being a natural exploration candidate. Memory-Efficient DQN [33] reports on 5 of 57 games selected for memory-sensitivity, leaving the method's broader performance profile undefined. The dashes are honest about reporting scope; reading them as evidence is internally consistent with the authors' own treatment of their methods.

F. The "improvement ladder" and its limits: Within several axes — particularly W1 (overestimation) and W4 (credit assignment) — a consistent chronological improvement trajectory is visible: classical DQN → Double DQN → Dueling / Prioritized → Distributional / Quantile-based methods, with each generation outperforming the previous on its diagnostic category. This "ladder" is real but axis-specific. A method that places at the top of the W4 ladder (FQF) does not necessarily improve on W3 (FQF scores zero on Montezuma). The ladder is a strong frame *within an axis* and a weak frame *across axes* — which is the central empirical argument for the structural pivot.

G. Why we do not produce a leaderboard: Earlier surveys [8–10] bold or italicize the "best" result per game. This presentation implicitly invites comparison across methods that used different evaluation protocols (training-frame budgets, seed counts, no-op-start variations, sticky-action settings). Boldface in the presence of protocol divergence overstates the strength of comparisons. We refrain from highlighting per-cell best-results in Tables II/III; the axis-stratified narrative above

TABLE XI
REPORTED ATARI BENCHMARK PERFORMANCE (RAW PER-GAME SCORES), PART I. REACTION-TIME CONTROL / STRATEGIC PLANNING / SPARSE REWARDS / DENSE REWARDS. "—" INDICATES THE ORIGINAL PAPER DID NOT REPORT A SCORE FOR THAT GAME.

Method (year)	Method type	Breakout	Sp. Inv.	Ms. Pac-Man	Q*bert	Montezuma	Pitfall!	Boxing	Enduro
Param Space Noise (2017)	Statistical	390	1,205	—	7,525	0	-100	—	1,672
C51 (2017)	Statistical	748	5,747	3,415	23,784	0	0	98	3,454
NoisyNet (2018)	Statistical	516	2,186	2,722	15,545	3	0	89	1,240
QR-DQN (2018)	Statistical	742	20,972	5,821	572,510	0	0	100	2,355
IQN (2018)	Statistical	734	28,888	6,349	25,750	0	0	100	2,359
FQF (2019)	Statistical	854	140	7,632	27,524	0	0	98	2,371
Nature DQN (2015)	Q-Func. Comp.	401	1,976	2,311	10,596	0	—	72	302
Deep Recurrent Q (2015)	Q-Func. Comp.	—	—	2,048	—	—	—	—	—
Double DQN (2016)	Q-Func. Comp.	375	3,155	3,210	14,875	0	—	82	320
Dueling DQN (2016)	Q-Func. Comp.	345	6,427	6,284	19,220	0	0	99	2,258
Rainbow DQN (2018)	Q-Func. Comp.	418	18,789	5,380	33,818	384	0	100	2,126
CBDQ (2025)	Q-Func. Comp.	—	—	—	—	—	—	—	—
DQN (2013)	Memory/Replay	168	581	—	1,952	—	—	—	470
Prioritized ER (2016)	Memory/Replay	481	1,697	965	12,741	44	-194	70	1,266
DQfD (2018)	Memory/Replay	308	—	4,696	21,793	4,638	57	99	2,200
MeDQN (2023)	Memory/Replay	—	—	—	—	—	—	—	—
Bootstrapped DQN (2016)	Ensemble	855	2,893	2,983	15,093	100	—	93	1,591
UCB Q-Ensemble (2018)	Ensemble	411	2,627	3,425	14,198	4	-1	98	2,753
Ensemble Bootstrap-ping (2021)	Ensemble	406	—	—	14,384	—	—	—	—
Posterior Sampling DQN (2023)	Model-Based	46	511	1,824	4,245	0	-44	79	363
Parallel Q (PQN, 2024)	Pure Q	515	18,451	5,568	31,717	0	-89	100	2,349

substitutes evaluation against the *diagnostic categories* for the weaker comparison against per-game best.

Two recent methodological contributions inform this decision and should be read alongside Tables II/III. [Agarwal et al. 2021, *Deep Reinforcement Learning at the Edge of the Statistical Precipice*] documents how the field's standard reporting practice — mean scores over three to five seeds, no confidence intervals — systematically misrepresents algorithmic performance. They introduce the *rliable* framework, which substitutes robust aggregate metrics (interquartile mean, optimality gap, probability of improvement) and stratified bootstrap confidence intervals for the point-estimate comparisons that have dominated Atari reporting since DQN [1]. Where the original-paper numbers we extract in Tables II/III are point-estimate scores under heterogeneous protocols, rliable provides the standard against which future Q-learning reports should be held. [Castro et al. 2020, *Revisiting Rainbow: Promoting More Insightful and Inclusive Deep Reinforcement Learning Research*] makes a complementary methodological case: small-scale, insight-oriented evaluation on the four-game ALE subset chosen to exercise distinct algorithmic dimensions can outperform full 57-game sweeps for understanding *which* mechanism in a compound agent matters. Castro et al.'s framework is methodologically aligned with the axis-

TABLE XII
Reported Atari benchmark performance, Part II. Large Observation Space / Partially Observable / Stochastic Environments.

Method (year)	Method type	River Raid	Priv. Eye	Frostbite	Hero	Zaxxon	Berzerk
Param Space Noise (2017)	Statistical	—	100	1,310	—	8,050	—
C51 (2017)	Statistical	17,322	15,095	3,965	38,874	10,513	1,645
NoisyNet (2018)	Statistical	9,425	3,712	753	6,246	6,920	905
QR-DQN (2018)	Statistical	17,571	350	4,384	21,395	13,112	3,117
IQN (2018)	Statistical	17,765	200	4,324	28,386	21,772	1,053
FQF (2019)	Statistical	23,561	140	16,473	30,926	15,180	12,422
Nature DQN (2015)	Q-Func. Comp.	8,316	1,788	328	19,950	4,977	—
Double DQN (2016)	Q-Func. Comp.	12,015	670	242	20,357	10,182	—
Dueling DQN (2016)	Q-Func. Comp.	21,163	103	4,673	20,818	13,886	3,409
Rainbow DQN (2018)	Q-Func. Comp.	—	4,234	9,591	55,887	22,210	2,546
Prioritized ER (2016)	Memory/Replay	10,206	2,202	289	15,151	9,501	644
DQfD (2018)	Memory/Replay	18,735	42,457	—	105,929	—	—
Bootstrapped DQN (2016)	Ensemble	12,845	1,813	2,181	21,021	11,492	—
UCB Q-Ensemble (2018)	Ensemble	15,622	1,252	1,903	—	3,695	—
Ensemble Bootstrapping (2021)	Ensemble	—	100	—	—	—	—
Posterior Sampling DQN (2023)	Model-Based	3,858	68	929	7,965	4,413	386
Parallel Q (PQN, 2024)	Pure Q	28,764	100	7,314	26,099	23,538	18,542

Task category	Diagnoses weakness(es)
Reaction-Time Control	function-approx stability (W8), sample inefficiency (W2)
Strategic Planning	reward sparsity & credit assignment (W4)
Sparse Rewards	brittle exploration (W3)
Dense Rewards	none specifically — baseline for stability
Large Observation Space	function-approx stability (W8)
Partially Observable	slow adaptation / recurrence (W7)
Stochastic Environments	overestimation bias (W1), distribution robustness

stratified evaluation philosophy of this paper: their "insight-oriented" dimensions and our weakness axes both substitute mechanism-targeted evidence for breadth-by-default reporting. The limitations we discuss in §V.H below should be read in this context: not as Atari is bad, but as Atari is one evidence stream among several, and the conventions for reporting Atari results have themselves been updated since the period covered by Tables II/III.

H. Limitations of Atari as a Q-learning benchmark: The preceding subsections treat Atari as the evidence stream against which methods are evaluated; this subsection turns to what Atari *cannot* evaluate. Six inherent limits constrain the conclusions drawable from Tables II/III:

H.1. Determinism. The Arcade Learning Environment [2] is deterministic by default: identical action sequences from identical initial frames produce identical trajectories. Two partial mitigations have become standard — *no-op starts* (the agent skips a random number of frames at episode start) and *sticky actions* [83] (actions persist with probability 0.25 per frame). Neither restores the stochasticity of real-world environments. Methods that exploit deterministic transitions — frame-perfect action sequences, memorized trajectories — score artificially well. The reporting protocols of methods covered here vary on this point; comparability suffers as a result.

H.2. Discrete action space. Atari games have 4–18 discrete actions per game. Continuous-control evaluation falls entirely outside Atari's scope. This excludes a significant fraction of modern Q-learning research from Atari-based comparison: REDQ (§IV.A), soft Q-learning, and the offline-RL methods of §IV.E are all evaluated primarily on MuJoCo / D4RL, not Atari. Cross-paper comparison across the discrete/continuous boundary is structurally limited.

H.3. Single-task per episode. Each Atari episode is one game; the agent does not transfer mid-episode between tasks. Multi-task, meta-learning, and continual-RL capabilities (§IV.G) cannot be evaluated on within-distribution Atari without modification. Agent57's achievement is impressive precisely because it handles the *distribution* of 57 games with a single agent — but the distribution is itself the construction; Atari was not designed as a multi-task benchmark.

H.4. Fixed environments. Every Atari game has a fixed level layout, a fixed sprite set, fixed game mechanics. There is no procedural variation, no held-out test environment, no OOD evaluation possibility. A Q-function that overfits to the training trajectories of Pong cannot be distinguished, on Atari alone, from one that learned the underlying game dynamics. Reports of human-level performance on Atari elide this distinction.

H.5. Compute scale. Agent57's 78B-frame training run is inaccessible to most research labs. R2D2's 10B is similar. The median academic Atari result of the past five years has been trained at 200M–1B frames — within an order of magnitude of Rainbow in 2017. The compute frontier of Atari research has effectively moved beyond what most researchers can reproduce, raising the question of whether Atari rankings produced at lower compute budgets remain comparable to frontier results.

H.6. Visual-only observation. Atari observations are 84×84 grayscale image patches. No language, no proprioception, no audio. Modern RL increasingly couples Q-learning with language-model priors (instruction following, language-conditioned exploration); Atari evaluation cannot speak to these directions.

These limits define a specific evaluation niche. Atari is a strong benchmark for *exploration under sparse reward* (§IV.C), *credit assignment in long-horizon discrete tasks* (§IV.D), and *stability of deep Q-learning at scale* (§IV.H). It is a weak benchmark for *sample efficiency in low-compute regimes*, *generalization across procedural variation*, *transfer across tasks*, and *robustness in deployment*. The choices to add coverage of D4RL in §IV.E and SMAC in §IV.F reflect which Atari-uncovered weaknesses require their own benchmark infrastructure to evaluate; the next subsection surveys further benchmarks that complement Atari along the dimensions above.

I. Newer benchmarks for Q-learning evaluation: The benchmarks below address one or more of the limits identified in §V.H. Each is tied to the axis it most directly diagnoses:

Atari-100k [84] caps training at 100,000 environment steps — roughly 2 hours of game play, 1/2000 the compute of standard Atari. The benchmark exposes *sample efficiency* under tight budgets, separating algorithmic improvements from compute scaling. Methods that perform well on standard Atari but poorly on Atari-100k (notably Rainbow without modifications) demonstrate that their gains rely on compute access. Primary axis: §IV.B (sample inefficiency).

ALE-stochastic. The Machado et al. (2018) revisions to the Arcade Learning Environment introduce sticky actions as the default and require evaluation on multiple difficulty modes. The revised protocol is increasingly the standard for new methods. Primary axis: §IV.H (stability under stochasticity).

ProcGen [85] provides 16 procedurally-generated games with disjoint training and evaluation level distributions. Methods are scored on held-out levels not seen during training, directly measuring generalization across procedural variation. Q-learning baselines on ProcGen show *very poor* generalization — between 25% and 50% of training performance on the held-out distribution — exposing a capability that Atari cannot diagnose. Primary axis: cross-axis (§IV.E distribution shift in the deployment direction; §IV.B sample efficiency on out-of-distribution data).

NetHack [86] provides a single procedurally- generated environment with millions-of-step episodes, rich symbolic observations, and extreme stochasticity. Q-learning baselines on NetHack score within an order of magnitude of random play, even at scale. The benchmark exposes the interaction of *credit assignment* (§IV.D) and *exploration* (§IV.C) at horizons orders of magnitude longer than Atari, and is currently the field's strongest test of whether Q-learning can scale to genuinely long-horizon decision- making.

BSuite [87] is DeepMind's *behavior suite for reinforcement learning*: twenty small experiments each designed to isolate a single capability — basic memory, generalization, noise robustness, scale, exploration, credit assignment, and others. BSuite is the closest existing benchmark to the axis-stratified evaluation philosophy of this paper. A method's BSuite profile directly maps to its position on the eight weakness axes; the benchmark's results report uses a radar-chart presentation that makes axis-level comparison visible at a glance. Primary axis: all eight (BSuite is meta-axial by design).

D4RL [88] is covered in §IV.E as the dominant offline-RL benchmark. Its locomotion, AntMaze, Adroit, and Kitchen suites diagnose *distribution shift* (§IV.E) along five distinct behavior-policy regimes.

SMAC [89] is covered in §IV.F as the cooperative multi-agent benchmark. Its scenarios diagnose *multi-agent coordination* (§IV.F) at varying levels of joint-task difficulty.

Beyond these, several benchmarks are emerging as Q-learning evaluation targets but lie outside this paper's scope: MetaWorld [90] and Meta-MuJoCo for meta-learning (§IV.G); RLBench [91] for robotic manipulation; Crafter [92] as a long-horizon survival benchmark with explicit achievement metrics; CARL [93] for context- generalization in continuous control. The expansion of benchmark diversity in the past five years is itself a contribution of the modern era: where the original DQN paper [18] was evaluated on seven Atari games, today's serious Q-learning methods typically report on three or more benchmark families.

VI. Empirical Evaluation of Tabular Q-Learning Variants

To complement the literature analysis of §IV and the Atari extraction of §V, we conduct controlled experiments in tabular environments where true dynamics are known and function approximation is unnecessary. This setup serves a specific methodological purpose: **isolating algorithmic design from architectural confound.** Many deep-RL claims about overestimation (W1), exploration (W3), or stability (W8) are conflated in evaluation by the presence of neural-network function approximation, replay buffers, and target networks. Tabular evaluation removes these confounds and exposes the algorithmic core.

A. Experimental setup: We implement each algorithm from scratch and evaluate on three canonical environments from the Gymnasium suite:

TABLE XIII
FINAL PERFORMANCE ON FROZENLAKE-V1 (MEAN ± STD OVER THE LAST 10 EVALUATION EPISODES, 5 SEEDS).

Algorithm	Final Reward
CVPI	1.00 ± 0.00
Double Q-Learning	0.90 ± 0.30
Expected SARSA	0.94 ± 0.24
MCTS	0.00 ± 0.00
MPI	1.00 ± 0.00
Multi-Step Q-Learning	0.90 ± 0.30
Q-Learning	0.96 ± 0.20
SARSA	0.78 ± 0.41
VI	0.00 ± 0.00

TABLE XIV
FINAL PERFORMANCE ON TAXI-V3 (MEAN ± STD OVER THE LAST 10 EVALUATION EPISODES, 5 SEEDS).

Algorithm	Final Reward
CVPI	5.52 ± 15.22
Double Q-Learning	-105.54 ± 44.62
Expected SARSA	-44.18 ± 41.00
MCTS	-6.22 ± 4.44
MPI	-0.90 ± 29.35
Multi-Step Q-Learning	-57.82 ± 80.61
Q-Learning	-57.04 ± 43.23
SARSA	-48.36 ± 45.84
VI	7.58 ± 2.74

- **FrozenLake-v1** — a stochastic gridworld with sparse terminal rewards. Tests exploration (W3) and overestimation under stochasticity (W1).
- **Taxi-v3** — a discrete planning task with structured rewards and a long horizon. Tests credit assignment (W4) and sample efficiency (W2).
- **CliffWalking-v1** — a deterministic gridworld with a high-penalty hazard along the shortest path. Tests the on-policy vs. off-policy trade-off (SARSA vs. Q-learning) and the conservatism induced by Expected SARSA.

Performance is measured by the average reward over the final 10 evaluation episodes, averaged across 5 random seeds. The reported values are raw — no smoothing is applied.

Two algorithms from the §IV survey are excluded from tabular evaluation. Bayesian Q-learning [42] is excluded because its posterior maintenance is prohibitively slow in repeated runs; its formal contribution is preserved in the §IV review. Neural Fitted Q Iteration (NFQ) [94] is excluded because it requires function approximation by construction.

B. Results: Tables IV, V, and VI report final-episode performance per environment, averaged across seeds.

C. Interpretation by axis: The tabular results admit a cleaner axis-attribution than Atari results because the algorithmic mechanism is the dominant source of performance variation.

Foundational baselines (Q-Learning, SARSA, Expected SARSA). All three achieve near-optimal performance on FrozenLake (Q-Learning 0.96, SARSA 0.78, Expected SARSA 0.94). Q-Learning's slight edge over SARSA reflects the off-policy advantage when the optimal policy is deterministic; SARSA's deficit reflects its on-policy update being biased toward the exploratory ϵ-greedy policy. On Cliff- Walking — where the on-policy/off-policy distinction is sharpest — SARSA's conservative behavior (-21) outperforms Q-Learning's optimistic one (-39), as expected: SARSA learns the policy actually being executed, while Q-Learning learns a riskier optimal policy that ε-greedy execution sometimes plunges off the cliff.

TABLE XV
FINAL PERFORMANCE ON CLIFFWALKING-V1 (MEAN ± STD OVER THE LAST 10 EVALUATION EPISODES, 5 SEEDS).

Algorithm	Final Reward
CVPI	-13.00 ± 0.00
Double Q-Learning	-53.38 ± 85.45
Expected SARSA	-25.02 ± 23.83
MCTS	-1.00 ± 0.00
MPI	-13.00 ± 0.00
Multi-Step Q-Learning	-36.94 ± 47.87
Q-Learning	-39.42 ± 49.55
SARSA	-21.02 ± 13.93
VI	-13.00 ± 0.00

Multi-Step Q-Learning (§IV.D). Multi-step achieves 0.90 on FrozenLake and -57 on Taxi, slightly below baseline. The result is informative about the limits of n-step in pure tabular settings: without function approximation, the variance penalty of larger n is not offset by the credit-assignment benefit. The empirical benefits of n-step learning observed in deep RL (e.g., as a Rainbow component) emerge specifically from the interaction with function approximation.

Double Q-Learning (§IV.A). Double Q-Learning performs comparably to Q-Learning on FrozenLake (0.90) but degrades substantially on Taxi (-105.5). The Taxi result is consistent with the §IV.A trade-off discussion: Double Q-Learning trades over-estimation for under-estimation, and on tasks where the optimal policy requires optimism about long-horizon rewards (Taxi has a -1 step penalty incentivizing greedy long-horizon action), under-estimation hurts. This is exactly the regime that the bias-variance frontier discussion of §IV.A.E flags as underexplored.

Planning baselines (Value Iteration, Policy Iteration, MPI, CVPI). Value Iteration and Policy Iteration are not RL algorithms in the strict sense — they require known dynamics — but serve as optimality references. CVPI [95] consistently matches or exceeds the RL methods, confirming that the gap between learned and optimal policies is the cost paid for unknown dynamics. The comparison quantifies the cost: on Taxi, CVPI scores 5.5 against Q-Learning's -57.

MCTS. MCTS underperforms on Taxi (-6.2) and FrozenLake (0.00) in the limited-budget regime evaluated. The result reinforces a point visible throughout this paper: planning methods scale to small state spaces in laboratory settings but degrade rapidly when the search budget is constrained relative to environment complexity.

D. Why tabular matters for the axis argument: The tabular results matter because they *test the axis attributions of §IV* in a setting where confounds are minimized. Double Q-Learning's tabular under-estimation pattern, visible on Taxi, is the same mechanism that drives §IV.A's bias-variance discussion in the deep setting — but in the tabular setting it can be observed cleanly. The convergence of tabular and deep evidence on the same mechanistic claims, when present, is one of the stronger methodological supports for the problem-first organization.

E. Statistical reporting and limitations: The five-seed mean ± standard-deviation reporting of Tables IV–VI is the standard convention for tabular Q-learning studies but is itself insufficient by the standards of [Agarwal et al. 2021, *Deep RL at the Edge of the Statistical Precipice*], which demonstrates that endpoint-only point-estimate comparisons substantially overstate the strength of conclusions drawn from small-seed RL experiments. We treat the §VI numbers as *indicative* of mechanism-driven trends rather than as statistically conclusive comparisons. A reproducibility-grade follow-up would adopt rliable's robust aggregate metrics (interquartile mean and probability of improvement with stratified bootstrap confidence intervals) over at least twenty seeds per algorithm-environment pair, and would publish learning curves alongside endpoint scores. We mark this as a methodological limitation rather than fixing it inside this section: the tabular results in §VI are not the load-bearing empirical evidence for the §IV axis claims (the Atari extraction of §V and the per-axis empirical-evidence subsections of §IV play that role); §VI primarily isolates algorithmic mechanism from architectural confound on small environments and is useful in proportion to that limited goal.

The MCTS and Value/Policy/Modified-Policy/Combined-Value-Policy Iteration entries in Tables IV–VI deserve a framing note: they are *planning oracles*, not learning peers. Value Iteration (VI), Policy Iteration (PI), Modified Policy Iteration (MPI), and Combined Value-Policy Iteration (CVPI) require known transition dynamics and reward functions, which the Q-learning methods discover by interaction. Including them in the same table simplifies presentation but invites a comparison they were not designed for. Readers should interpret the planning-method entries as performance upper bounds reachable when dynamics are fully known, and the gap between learning methods and planning oracles as the cost of solving the unknown-dynamics problem.

VII. Comparative Analysis of Deep Q-Learning Repositories

This section analyzes algorithmic coverage of Q-learning variants across six widely-used open-source deep RL repositories — Tianshou [12], XuanCe [13], CleanRL [14], DQN Zoo [15], Stable Baselines3 [16], and RLlib [17] — together with their stated design priorities. Table VII reports a method-level coverage matrix; Table VIII summarizes design trade-offs.

A. Scope of this analysis: This paper's repository analysis is *taxonomic*: we report which algorithms each repository implements, with brief commentary on the design choices that shape coverage. We do *not* benchmark the repositories' implementations against one another; that question is addressed empirically by Hundal et al. [96], whose controlled comparison of PPO across five repositories shows substantial reproducibility variance even for a single algorithm under nominally equivalent configurations. Hundal et al.'s work sits in a broader reproducibility-crisis literature: Henderson et al. 2018 [*Deep Reinforcement Learning That Matters*] showed that hyperparameter sensitivity and seed variance in deep RL can rival the gap between published baselines and proposed methods; Engstrom et al. 2020 attributed several headline performance claims in policy-gradient methods to implementation details rather than algorithmic substance. Hundal et al.'s empirical-reproducibility audit and the taxonomic-coverage analysis here are complementary: practitioners selecting a repository care both *whether their algorithm is supported* (this paper's question) and *whether the implementation reproduces published results* (Hundal et al.'s question, against the standards of the reproducibility-crisis literature).

Audit methodology and date. The repository coverage matrix in Table VII reflects the state of each repository as of 2026-05-01. Specifically, the audited versions were Tianshou 0.5.x, XuanCe 1.x, CleanRL master (commit-pinned during the audit), DQN Zoo (current head), Stable Baselines3 2.x, and RLlib 2.x. Repository state changes quickly; readers consulting this paper after 2026 should treat Table VII as a snapshot. We have not pinned commit hashes for each cell in the matrix because the matrix is intended to characterize *general availability* of algorithms rather than reproducibility-grade implementation fidelity; for the latter purpose, audit-with-commit-hash methodology in the style of Hundal et al. 2025 is the appropriate framework.

Named-algorithm support vs. feature-equivalent support. The matrix marks a method as *supported* if the repository implements it as a named algorithm with appropriate hyperparameter exposure. This is the strictest interpretation of "supported" and may *under-state* practical availability in two ways. First, several repositories expose component features that can be combined to reproduce a method without naming it: RLlib's `DQNConfig`, for example, accepts flags for `noisy: True`, `categorical: distributional_n_atoms`, and similar that together can configure a Rainbow-like agent without the matrix counting RLlib as a named-Rainbow implementation. Second, some repositories maintain experimental branches or community PRs that implement recent methods but have not been merged to master; we excluded these from the matrix to keep the audit reproducible. The practical implication is that Table VII reflects what users will find by searching repository documentation for an algorithm by name; the *feature-equivalent* availability of compound methods (Rainbow components, ensemble variants, distributional configurations) is consistently higher than the matrix indicates, particularly in RLlib and to a lesser extent in Tianshou and XuanCe.

B. Axis-aware coverage: Reading Table VII through the eight axis-sections of §IV reveals patterns invisible in the method-type organization of the original matrix:

Well-supported axes. §IV.A (overestimation, via Double DQN / Dueling DQN) and §IV.D (distributional, via C51 / QR-DQN / IQN / FQF) are the best-covered axes across repositories. Tianshou alone implements all six method-level entries in these axes; XuanCe and RLlib follow. The bias toward overestimation- and distributional-method coverage reflects the empirical strength of these families during the 2016–2019 deep-RL boom, when most of the repositories were

Method	Axis	§IV reference
Deep Recurrent Q (DRQN)	W7, scaling/adaptation	§IV.G.B.4
Cognitive Belief-Driven Q (CBDQ)	W3, brittle exploration	§IV.C.B.3
DQfD	W2, sample inefficiency	§IV.B.B.2
Memory-Efficient DQN	W2, sample inefficiency	§IV.B.B.3
Bootstrapped DQN	W3, brittle exploration	§IV.C.B.2
UCB Q-Ensemble	W3, brittle exploration	§IV.C.B.2
Ensemble Bootstrapping (EBQL)	W1, overestimation	§IV.A.B.2
Posterior Sampling DQN	W3, brittle exploration	§IV.C.B.3
Parallel Q-Learning (PQN)	W8, function-approx stability	§IV.H.B.3

architected.

Patchily-supported axes. §IV.B (sample inefficiency) is covered unevenly: PER is in three of six repositories, MeDQN in zero, DQfD in zero, HER in three of six (but typically only as part of goal-conditioned baselines, not as a Q-learning enhancement). §IV.C (exploration) is similarly patchy: NoisyNet is in one of six repositories (RLlib), Bootstrapped DQN in zero, UCB Q-Ensemble in zero. The patchiness reflects the rapid evolution of these axes and the difficulty of integrating per-method changes into production-grade architectures.

Entirely-absent axes. §IV.F (multi-agent value decomposition) and §IV.G (distributed-scale methods) are functionally absent from the surveyed Q-learning repositories. RLlib supports multi-agent RL via its actor framework but does not implement VDN, QMIX, QPLEX, or QTRAN as native Q-learning methods. Ape-X, R2D2, and Agent57 have no implementations in any of the six repositories at the time of writing. The absence is structural: these methods require distributed infrastructure beyond the single-machine scope of most repositories.

§IV.E (offline RL) is in transition. None of the six surveyed repositories implement CQL, IQL, BCQ, or EDAC natively as of the draft's evaluation cutoff. Practitioners use d3rlpy, Clean-Offline-RL, or repository forks for these methods. The gap between mainstream-DRL repository coverage and the offline-RL practical landscape is one of the larger inconsistencies the axis-aware view exposes.

C. Methods absent from all six repositories: Nine methods covered in §IV are absent from *every* surveyed repository:

The list overlaps substantially with the methods identified in §IV as productive directions for future work, suggesting that *implementation availability* is itself a bottleneck on practical adoption of methodological advances. This observation motivates the future-work proposal in §VIII for a community-maintained Q-learning–specific repository.

D. Tables VII and VIII — coverage and design trade-offs: Reading Table VIII, the design priorities of each repository emerge:

- **Tianshou** and **XuanCe** prioritize coverage breadth across methods at the cost of single-method depth or reproducibility guarantees.
- **CleanRL** prioritizes single-file readability at the cost of modularity, which limits coverage of compositional methods (Rainbow, for instance, is harder to express in CleanRL's single-file architecture).
- **Stable Baselines3** prioritizes API stability and broad user adoption, with the cost of slower integration of new methods.
- **RLlib** prioritizes distributed scalability and production use, with the cost of complexity and difficulty in debugging individual algorithms.
- **DQN Zoo** prioritizes reproduction fidelity to DeepMind's published results, with the cost of restricted scope (DQN family only).

The design priorities are not in conflict but represent different positions on a Pareto frontier: a repository optimizing for one property necessarily relaxes another. The practitioner-side implication is that selection of a repository is itself an axis decision: a researcher exploring overestimation methods is well- served by Tianshou; a practitioner deploying at scale is well-served by RLlib; an educator demonstrating algorithms is well-served by CleanRL. No single repository serves the field optimally — and the axis-aware coverage gaps reported above suggest that no repository currently serves the Q-learning research community comprehensively.

E. Toward a Q-learning–specific repository: The axis-aware analysis above motivates a specific deliverable: **a community-maintained repository focused exclusively on Q-learning and its deep variants**, providing unified training scripts, standardized benchmarks, modular algorithm implementations, and shared logging tools. Such a repository would prioritize coverage of the methods identified in §VII.C as absent from all existing repositories, with empirical reproducibility audits in the style of Hundal et al. as a release-quality gate.

Section VIII discusses this proposal as the paper's principal forward-looking deliverable.

VIII. Conclusion and Future Work

A. Summary: This paper presents a problem-first survey of Q-learning and deep Q-learning, reorganizing three decades of methodological development around the eight foundational weaknesses of vanilla Q-learning introduced in §II.B: overestimation bias, sample inefficiency, brittle exploration, reward sparsity and credit assignment, distribution shift, multi-agent coordination, slow adaptation, and function-approximation instability.

For each weakness, §IV surveys the families of methods that respond:

- **§IV.A** — overestimation control via decoupling (Double Q, Double DQN) and ensembles (EBQL, REDQ).
- **§IV.B** — sample efficiency via prioritized sampling (PER), demonstration augmentation (DQfD), memory-efficient consolidation (MeDQN), and goal relabeling (HER).
- **§IV.C** — directed exploration via noise injection (NoisyNet, Parameter Space Noise), ensemble disagreement (Bootstrapped DQN, UCB Q-Ensemble), belief modulation (CBDQ, PSDQN), and intrinsic motivation (RND, Go-Explore).

TABLE XVI
REPOSITORY SUPPORT FOR DEEP Q-LEARNING ALGORITHMS, GROUPED BY CATEGORY. ● = IMPLEMENTED; ○ = NOT IMPLEMENTED.

Category	Method (year)	Tianshou	XuanCe	CleanRL	SB3	RLlib	DQN Zoo
Statistical	Param Space Noise (2017)	○	●	○	○	○	○
Statistical	C51 (2017)	●	●	●	○	●	●
Statistical	NoisyNet (2018)	○	○	○	○	●	○
Statistical	QR-DQN (2018)	●	●	○	●	○	●
Statistical	IQN (2018)	●	○	○	○	○	●
Statistical	FQF (2019)	●	○	○	○	○	○
Q-Func. Comp.	Nature DQN (2015)	●	●	●	●	●	●
Q-Func. Comp.	DRQN (2015)	○	○	○	○	○	○
Q-Func. Comp.	Double DQN (2016)	●	●	○	○	●	●
Q-Func. Comp.	Dueling DQN (2016)	●	●	○	●	●	○
Q-Func. Comp.	Rainbow DQN (2018)	●	○	○	○	●	●
Q-Func. Comp.	CBDQ (2025)	○	○	○	○	○	○
Memory/Replay	DQN (2013)	○	○	○	○	○	○
Memory/Replay	Prioritized ER (2016)	○	●	○	○	●	●
Memory/Replay	DQfD (2018)	○	○	○	○	○	○
Memory/Replay	MeDQN (2023)	○	○	○	○	○	○
Ensemble	Bootstrapped DQN (2016)	○	○	○	○	○	○
Ensemble	UCB Q-Ensemble (2018)	○	○	○	○	○	○
Ensemble	Ensemble Bootstrap-ping (2021)	○	○	○	○	○	○
Model-Based	Posterior Sampling DQN (2023)	○	○	○	○	○	○
Pure Q	Parallel Q (PQN, 2024)	○	○	○	○	○	○

- §IV.D — credit assignment via multi-step returns and the distributional family (C51, QR-DQN, IQN, FQF).
- §IV.E — offline RL via policy constraint (BCQ, AWAC, BRAC), value penalty (CQL), expectile regression (IQL), and ensemble diversification (EDAC).
- §IV.F — cooperative value decomposition via additive (VDN), monotonic mixing (QMIX), duplex dueling (QPLEX), and constraint- free (QTRAN) factorizations.
- §IV.G — scaling and adaptation via distributed actor-learner architectures (Ape-X, R2D2), bandit-controlled exploration meta-policies (Agent57), and meta-learning on the Q-function.
- §IV.H — function-approximation stability via target networks, architectural decomposition (Dueling), normalization recipes (PQN), and consolidation losses (MeDQN, Munchausen DQN).

Sections V and VI present axis-stratified empirical evidence, showing that methods targeting weakness W_i excel on the task categories that diagnose W_i and remain near baseline elsewhere. Section VII analyzes algorithmic coverage across

TABLE XVII
COMPARISON OF POPULAR DEEP RL REPOSITORIES — DESIGN PROS AND CONS.

Repository	Pros	Cons
Tianshou	Dual API (high-level and low-level); emphasis on reproducibility with agent-level tests; native TensorBoard support	Logs/results not available for all algorithms; reproducibility guarantees not empirically verified across full benchmark suite
XuanCe	Modular YAML config files; W&B integration for hyperparameter tuning; high modularity for code reuse	Default hyperparameters often diverge from original papers; incomplete support across environments; steep learning curve
CleanRL	Single-file implementations; easy to audit and understand; lightweight and minimal dependencies	Limited support for large-scale experiments; less modular, harder to extend
Stable Baselines3	Clean API with sklearn-style interface; well-maintained and widely adopted; compatible with VecEnv, Gym, etc.	Focused more on policy-gradient methods; limited Q-learning variants beyond DQN
RLlib	Distributed training out-of-the-box; Ray Tune integration; production-grade scalability	High complexity and heavyweight; harder to debug or customize individual components
DQN Zoo	Faithful reimplementation of DQN variants; reproducibility aligned with DeepMind practices; highly organized training logs and configs	Focused exclusively on DQN family; less beginner-friendly

six open-source repositories, identifying nine methods absent from all surveyed codebases — a gap that motivates the principal forward-looking deliverable of this paper.

B. A community Q-learning repository: The repository coverage analysis of §VII surfaces a concrete implementation gap: nine methods covered in §IV — DRQN, CBDQ, DQfD, MeDQN, Bootstrapped DQN, UCB Q-Ensemble, EBQL, Posterior Sampling DQN, and PQN — are absent from *every* surveyed open-source repository. Across the new axes introduced in §IV.E, §IV.F, and §IV.G, the gap is broader still: most offline-RL, multi-agent value-decomposition, and distributed-Q methods have no implementation in any of the six repositories analyzed.

We propose the development of a **dedicated, community-maintained repository focused exclusively on Q-learning and its deep variants**. The repository would have four design priorities:

1. **Axis-aware coverage** — implementations organized by the eight weakness axes, prioritizing methods absent from existing repositories, with a coverage matrix maintained as the public roadmap.
2. **Unified training scripts** — a common interface across methods permitting like-for-like comparison under matched evaluation protocols.
3. **Standardized benchmarks** — Atari, classic control, D4RL, and SMAC included as canonical benchmark suites, with reference results for each implemented method.
4. **Empirical reproducibility audits** — each method release gated on reproduction of the original paper's headline results, in the style of Hundal et al. [96].

The repository's initial roadmap would prioritize the nine methods in §VII.C as the first implementation targets, followed by the offline-RL family (CQL, IQL, EDAC), the multi-agent value- decomposition family (QMIX, QPLEX, VDN), and the distributed-Q family (Ape-X, R2D2, Agent57). Cross-references between the repository's documentation and this paper's §IV sections would let practitioners navigate from "I face problem X" (this paper's axis) to "available implementations addressing X" (the repository's coverage matrix).

C. Open research directions, by axis: The "Open Questions" subsections (E) of each §IV axis-section collectively summarize the field's frontier. We highlight four cross-axis directions of particular promise:

The compute-vs.-algorithmic tradeoff. The empirical pattern in §V suggests that distributed scaling (§IV.G) accounts for the largest single performance increase in deep RL since DQN, but PQN's single-machine results suggest the algorithmic axes are far from saturated. A systematic study of which weaknesses are *compute-resoluble* versus *algorithmically-resoluble* — at fixed compute budgets — would clarify where research effort is best directed.

Cross-axis composition. Frontier agents (Agent57, NGU) explicitly compose mechanisms from multiple axes. A formal account of which axis combinations are synergistic, redundant, or antagonistic would convert the empirical pattern into a design principle. The Rainbow ablation [6] provides one such account on a smaller method set; a broader cross-axis composition study would extend it.

The offline-online interface. Methods bridging offline and online learning (AWAC, Cal-QL) address a regime increasingly central to practical deployment but theoretically underexplored. The interface is itself a candidate weakness axis the field may recognize in coming years.

Q-learning under real-world distribution shift. §IV.E treats distribution shift as a *training* regime: the data was collected by one or more behavior policies and is then fixed. A deployment-side analog — the learned policy meets a state distribution different from the one it was trained on — is largely unaddressed. Sim-to-real transfer in robotic Q-learning, OOD robustness against perturbations, and online policy correction at deployment time all live in this regime. The newer benchmarks of §V.I (ProcGen for procedural variation, CARL for context generalization, RLBench for sim-to-real) begin to provide evaluation infrastructure, but no unified Q-learning treatment of *deployment* distribution shift exists. This is the axis where the gap between Atari-evaluated methods and real-world deployment is most visible, and where Q-learning's value-based perspective could productively engage with robotics, control, and operations-research literature that has developed parallel tools.

D. Limitations and Ethics: This is a survey rather than a deployed system, so the relevant considerations are scholarly and community-facing rather than direct model-harm concerns. We surface five limitations explicitly.

Benchmark monoculture. Tables II–III's reliance on Atari and §VI's tabular Gymnasium environments reflects the field's historical convention. As documented in §V.H, Atari is a strong benchmark for a specific set of weaknesses (exploration, credit assignment, function-approximation stability) and a weak one for others (sample efficiency at small budgets, generalization across procedural variation, continuous control, real-world deployment). A survey that elevates Atari evidence to load-bearing status inherits these gaps. We have partly mitigated this through §V.I's coverage of ProcGen, NetHack, BSuite, D4RL, and SMAC and through §IV.E and §IV.F's adoption of D4RL and SMAC respectively, but the breadth of evaluation evidence remains uneven across the eight axes.

Compute inequality. Several of the methods covered (Agent57's 78B training frames, R2D2's 10B, large in-context Q-learning agents) require compute budgets accessible to only a handful of industrial labs. By including these methods alongside academically reproducible alternatives without flagging the gap, a survey risks implicitly endorsing a research culture where frontier work is structurally inaccessible to most readers. §IV.G's discussion of the compute-versus-algorithmic trade-off and §V.H.5 on compute scale begin to address this, but the field-wide tension between *algorithmic discovery* and *compute-driven discovery* is underdeveloped here and warrants explicit treatment in future surveys.

Citation bias in narrative reviews. Any narrative survey that reduces a large field to "canonical methods" can inadvertently erase negative results, replication studies, and less-publicized alternatives — even when, as here, the source selection criteria are documented (§III.A). We provide the prior-art sweep and a full per-method method-type index as supplementary material — the audit trail for our differentiation claim and a bridge to the older organization — but we cannot eliminate the bias inherent in narrative selection. Readers seeking a more systematic review-of-reviews should consult the Springer NCAA 2026 offline-RL distribution-shift survey and the broader-RL surveys of Ghasemi 2024/2025 and Murphy 2024/2025 as complementary navigations of overlapping terrain.

Stale tooling comparisons. The repository comparison of §VII is pinned to repository state as of early 2026. Implementation landscapes shift quickly; readers consulting this paper in 2027 or later should treat Table VII as a snapshot rather than current ground truth. The proposed Q-learning–specific repository (§VIII.B) is intended in part as a mechanism for maintained tooling comparison rather than as a one-shot artifact.

Frontier methods overstated relative to settled techniques. This paper devotes substantial space to in-context Q-learning (§IV.G.B.3), flow-matching policies (§IV.E.B.5), and recent theoretical advances (§IV.I.C) — all subject to ongoing community consensus formation. Some claims in those subsections may be revised as the field consolidates. We have flagged the relevant sections with explicit "active research area" language where applicable, but readers should treat any specific claim about post-2024 work as more uncertain than the well-established results in §IV.A–D and §IV.H.

A small note on broader ethical considerations: Q-learning's expanding role in robotics, recommender systems, and language-model alignment carries downstream policy implications that this survey does not address. A reader interested in the responsible deployment of value-based RL methods should consult the broader RL-safety and RL-policy literature (Saunders et al. 2023, Anderljung et al. 2024) as complementary material.

E. Closing: Q-learning's longevity as a foundational paradigm rests on its combination of conceptual simplicity and methodological extensibility. The thirty-five years since [Watkins 1989] have produced a tooling ecosystem rich enough to deserve organized treatment but specialized enough that no general-RL survey can capture its texture. This paper offers the first multi-axis problem-first synthesis of that ecosystem, paired with empirical and implementation analyses that support the structural claim.

The methodological landscape this paper traces is not closed. New axes will emerge — the offline-online interface and the meta-RL boundary are two candidates — and existing axes will continue to admit new mechanisms. The problem-first organization is intended to be extensible: new methods slot into existing axes, and new axes can be added as the field recognizes them. The framework's value is precisely that it survives those additions.

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